

Scale-invariant multipole local expansion features and a self-consistent exchange-hole functional

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[Abstract handled separately.] We introduce SIMPLE, a complete, orthonormal, scale-invariant local expansion of the electron density, and use it to build a self-consistent exchange functional from an explicit model of the exchange hole. The descriptors reproduce the homogeneous-electron-gas limit by construction, are exactly rotationally covariant, and remain numerically stable as the density vanishes. Projecting the Coulomb interaction onto the same orthonormal basis turns the exchange-energy integral into a closed contraction of monopole descriptors, yielding a functional that is exact in the homogeneous and one-electron (Fermi–Amaldi) limits and matches the gradient expansion to second order. Self-consistent atomic calculations reproduce optimized-effective-potential exchange potentials and Hartree–Fock energies across the first rows of the periodic table.

[Introduction handled separately.]

I. THEORY AND METHODS

A. SIMPLE feature definitions

The SIMPLE features are a *scale-invariant multipole local expansion* of the electron density — a local projection onto an orthonormal radial–angular basis, reconstructed at a density-adaptive radius so the coefficients are dimensionless and invariant under uniform scaling.

1. Projection, orthogonality, and local scale

Let ρ_σ be the spin- σ density. About a point $\mathbf{r}_0$ we project the local density onto an orthonormal kernel basis on the ball $B_{R_c} = \{\mathbf{r} : |\mathbf{r}| \leq R_c\}$,

$$c_{n\ell m}(\mathbf{r}_0) = \int_{B_{R_c}} K_{n\ell m}(\mathbf{r}) \rho_\sigma(\mathbf{r}_0 + \mathbf{r}) d^3r. \quad (1)$$

The kernel is $K_{n\ell m}(\mathbf{r}) = R_{n\ell}(r) Y_{\ell m}^{\mathbb{R}}(\hat{\mathbf{r}})$, with real spherical harmonics $Y_{\ell m}^{\mathbb{R}}$ and Dirichlet spherical-Bessel radial functions $R_{n\ell}(r) = \mathcal{N}_{n\ell} j_\ell(z_{n\ell} r/R_c)$ ($z_{n\ell}$ the positive zeros of j_ℓ), orthonormal under $\int_0^{R_c} r^2 R_{n\ell} R_{n'\ell} dr = \delta_{nn'}$; the $c_{n\ell m}$ are therefore the orthonormal expansion coefficients of the local density inside the ball. This is the same class of local density expansion used by atomistic descriptors such as SOAP [1], ACE [2], and SNAP [3], and by convolutional density functionals [4–6].

Local density scale. The local scale is the ball-averaged density and its Fermi wavevector, $\bar{\rho}_\sigma(R) = \frac{3}{R^3} \int_0^R r^2 \rho_{\sigma,00} dr$ and $\bar{k}_F(R) = (6\pi^2 \bar{\rho}_{\sigma,\text{safe}})^{1/3}$, where $\rho_{\sigma,00}$ is the spherical-average ($\ell = 0$) profile of the local density and the smooth floor $\bar{\rho}_{\sigma,\text{safe}} = (\bar{\rho}_\sigma^2 + \rho_{\min}^2)^{1/2}$ regularizes vanishing density ($\rho_{\min}$ a small density floor). Using the ball average rather than the point value is what keeps the features numerically stable as $\rho \rightarrow 0$.

Adaptive radius and the local Wigner–Seitz scale. A functional must see the same dimensionless environment regardless of the local density scale, so the expansion is reconstructed not at the fixed window R_c but at a density-adaptive radius R_{ad} that holds a fixed dimensionless cutoff ξ^* ,

$$g(R_{\text{ad}}) = \xi^*, \quad g(R) \equiv R \bar{k}_F(R), \quad R_{\text{ad}} \leq R_c \text{ (clamped)}. \quad (2)$$

Because $R_{\text{ad}} \bar{k}_F = \xi^*$ fixes the enclosed charge $([R_{\text{ad}} \bar{k}_F]^3 = 18\pi^2 M(R_{\text{ad}}))$, with the enclosed monopole moment $M(R) = \int_0^R r^2 \rho_{\sigma,00} dr$, R_{ad} is the local analogue of the Wigner–Seitz radius: the radius enclosing a fixed amount of charge measured in local Fermi wavelengths. In practice it is located by a single fixed-stencil “charge within a radius” convolution $M(R)$ and a closed-form inversion of $M(R_{\text{ad}}) = M^*$ (App. B), with no iterative solve. R_{ad} is a functional of the density alone and maps exactly to R_{ad}/λ under $\rho_\sigma(\mathbf{r}) \mapsto \lambda^3 \rho_\sigma(\lambda \mathbf{r})$.

2. Scale invariance and non-dimensionalization

We require features invariant under that uniform scaling. Exact invariance demands the adaptive cutoff $R_{\text{ad}} \propto \bar{k}_F^{-1}$ of Eq. (2), which cannot be a single fixed convolution stencil; SIMPLE keeps a fixed window and reconstructs the adaptive-window coefficients by a linear map that is exact when the adaptive radius lies below the cutoff and decays smoothly once R_{ad} would exceed it.

Dilation identity. A change of variables in Eq. (1) gives the exact covariance of the coefficients at corresponding cutoffs,

$$c_{n\ell m}[\rho_\lambda; R/\lambda] = \lambda^{3/2} c_{n\ell m}[\rho; R], \quad (3)$$

because the basis depends on r only through the ratio $z_{n\ell} r/R$ (App. B). A uniform scaling thus dilates the radial sampling wavevectors but loses no information: the descriptors an adaptive cutoff would produce are recoverable from fixed-window data.

Mean-split transfer. The reconstruction acts on the fluctuation about the window mean — the constant maps analytically through the monopole integrals $A_n(R) = \int_{B_R} K_{n00} d^3r$, removing boundary oscillations,

$$\tilde{c}_{n\ell m} = c_{n\ell m} - \bar{\rho}_\sigma(R_c) A_n(R_c) \delta_{\ell 0} \delta_{m 0}, \quad (4)$$

$$c'_{m\ell m'} = \bar{\rho}_\sigma(R_c) A_m(R_{\text{ad}}) \delta_{\ell 0} \delta_{m' 0} + \sum_{n=0}^{n_{\text{in}}-1} M_{mn}^{(\ell)}(R_{\text{ad}}) \tilde{c}_{n\ell m'}, \quad (5)$$

with transfer matrices $M_{mn}^{(\ell)}(R_{\text{ad}}) = \int_0^{R_{\text{ad}}} r^2 R_{m\ell}^{(R_{\text{ad}})} R_{n\ell}^{(R_c)} dr$. For a uniform density $\tilde{c} \equiv 0$, so the HEG limit is exact at any truncation. Surplus inner channels ($n_{\text{in}} > n_{\text{out}}$) supply the radial resolution the transfer consumes ($n_{\text{in}} \lesssim R_c/2h$ on a grid of spacing h).

Closed form. With $a = z_{m\ell}/R_{\text{ad}}$ and $b = z_{n\ell}/R_c$, the Sturm–Liouville cross-product identity evaluates the transfer matrices analytically, with no quadrature or tabulation,

$$\begin{aligned} & \int_0^R r^2 j_\ell(ar) j_\ell(br) dr \\ &= \frac{R^2 [a j'_\ell(aR) j_\ell(bR) - b j_\ell(aR) j'_\ell(bR)]}{b^2 - a^2}, \end{aligned} \quad (6)$$

which at $R = R_{\text{ad}}$ simplifies further since $j_\ell(aR_{\text{ad}}) = j_\ell(z_{m\ell}) = 0$ (App. B).

Non-dimensionalization. The SIMPLE features normalize the transferred coefficients by the local amplitude scale at the adaptive radius,

$$\varrho_{n\ell m}(\mathbf{r}_0) = \frac{c'_{n\ell m}(\mathbf{r}_0)}{A_0(R_{\text{ad}}) \bar{\rho}_{\sigma, \text{safe}}(R_{\text{ad}})}, \quad (7)$$

which is dimensionless for every ℓ and, by Eqs. (3) and (5), exactly invariant under uniform scaling in the continuum (App. C). The ball average gives a smooth $\rho \rightarrow 0$ limit, $\varrho_{n\ell m} = \mathcal{O}(\epsilon)$ and power spectrum $\sim \mathcal{O}(\epsilon^2)$ as $\rho \rightarrow \epsilon\rho$, with no overflow. After non-dimensionalization the homogeneous-gas limit is the fixed analytic vector $\varrho_{n00}^{\text{HEG}} = (-1)^n/(n+1)$, $\varrho_{n\ell m} = 0$ ($\ell > 0$): a uniform density carries only this monopole signature, and any departure from it measures inhomogeneity. The local scale k_F is retained as an explicit feature — the physically correct factorization, since exchange obeys $E_x[\rho_\lambda] = \lambda E_x[\rho]$ while correlation requires the scale.

The SIMPLE features $\varrho_{n\ell m}$ are thus a scale-free, orthonormal expansion of the density about a point — dimensionless, smoothly behaved in vacuum, and systematically improvable as R_c and the channel resolution increase.

B. Construction of SIMPLE invariants

Under a rotation $\mathcal{R}$ the features transform as rank- ℓ spherical tensors, $\varrho_{n\ell m} \mapsto \sum_{m'} D_{mm'}^{(\ell)}(\mathcal{R}) \varrho_{n\ell m'}$

(App. C): they are rotationally *covariant*, whereas a density functional needs inputs that are rotationally *invariant*. We construct invariants in three ways — the dimensionless reduced gradient and Laplacian, the power spectrum and bispectrum that form a complete basis, and targeted projections of specific physical operators.

1. Dimensionless gradient and Laplacian

The dimensionless inhomogeneity measures used by semilocal functionals — the reduced gradient $s = |\nabla\rho|/(2k_F\rho)$ and reduced Laplacian $q = \nabla^2\rho/(4k_F^2\rho)$ (with $k_F = (3\pi^2\rho)^{1/3}$) — are recovered directly from the SIMPLE features, with no fit or empirical decoder. Since $\nabla\rho$ is a vector (rank 1) and $\nabla^2\rho$ a scalar (rank 0), they are spectral contractions of the $\ell = 1$ and $\ell = 0$ channels,

$$s = c_s \sqrt{\sum_m \left(\sum_n g_n^{(1)} \varrho_{n1m} \right)^2}, \quad q = c_q \sum_n g_n^{(0)} \varrho_{n00}, \quad (8)$$

with fixed parameter-free spectral weights — $g_n^{(1)} \propto R'_{n1}(0)$ (the slope at the origin) and $g_n^{(0)} \propto k_n^2 R_{n0}(0)$ (the Bessel eigenvalue, $k_n = z_{n0}/R_c$) — and constants c_s, c_q (App. E). The non-dimensionalization already supplies the $1/(2k_F\rho)$ and $1/(4k_F^2\rho)$ factors, so s is simply the $\ell = 1$ power of the spectrally-weighted features and inherits their rotational and scale invariance.

The two ingredients differ sharply in practice. The reduced gradient is well-behaved, agreeing with direct finite differences to $\sim 1\%$ within the basis resolution (Sec. II A 3). The reduced Laplacian is recoverable by the same construction but *not numerically usable*: like a standard finite-difference Laplacian, the $\ell = 0$ eigenvalue sum weighted by k_n^2 amplifies high-wavevector noise, and its relative reconstruction error reaches tens to hundreds of percent in dense environments (Sec. II A 3, App. E). The reduced gradient is therefore a reliable ingredient and the Laplacian is not.

2. Power spectrum, bispectrum, and operator projections

Because the transfer acts on the radial index n while rotations act on the magnetic index m , the two commute, so any contraction of the $\varrho_{n\ell m}$ to total angular momentum zero is a rotational invariant (App. C). Two routes are useful. Generic invariants form a *complete basis* for rotationally invariant multilinear functionals of the density — the power spectrum $\tilde{P}_{n\ell} = \sum_m \varrho_{n\ell m}^2$, the bispectrum, and higher Clebsch–Gordan trees — systematically improvable as R_c and the resolution increase. This route is especially favorable for machine-learned functionals: rather than a fixed, hand-chosen descriptor set, the model is handed a complete, orthonormal, systematically convergent projection of the local environment, so its accuracy can be improved in a controlled way simply

by extending the basis. Alternatively one can *target a specific operator* whose symmetry is known and project it directly onto the features.

The decisive targeted example is the Coulomb operator $1/u$. Orthonormality converts any rotation-invariant ν -point kernel into a finite contraction $\sum T_{\{n_i \ell_i m_i\}}^{(\nu)} \prod_i \varrho_{n_i \ell_i m_i}$ with constant projected-kernel coefficients (App. D); for $1/u$ only the monopole ($\ell = 0$) average of the density survives the isotropic angular integral, $\bar{\rho}(r_0, u) = A_0(R_{\text{ad}}) \bar{\rho}_{\sigma, \text{safe}}(R_{\text{ad}}) \sum_n \varrho_{n00}(r_0) R_n(u)$, its monopole coefficients the dimensionless features ϱ_{n00} [Eq. (7)] scaled by the local amplitude $A_0 \bar{\rho}_{\sigma, \text{safe}}$ that restores the density units, and the operator is the radial Coulomb contraction

$$\int_{B_{R_c}} \frac{\rho(r_0 + \mathbf{u})}{u} d^3u = 4\pi \int_0^{R_c} \bar{\rho}(r_0, u) u du \quad (9)$$

$$= A_0 \bar{\rho}_{\sigma, \text{safe}} \sum_n C_n \varrho_{n00}(r_0),$$

with the basis Coulomb moments $C_n = 4\pi \int_0^{R_c} R_n(u) u du$. The $1/u$ weight reduces the volume measure $u^2 du$ to $u du$: the projection keeps the long-range $1/r$ dependence that semilocal functionals miss — it is *not* merely the average density. Contracted with the local density itself, Eq. (9) is the Hartree energy of the environment enclosed within R_c , so the Coulomb projection converts the density directly into an approximate local energy scale (used in Sec. II A 2 to calibrate R_c and n_{out}). More generally, projecting other physical operators onto the features in the same way supplies the building blocks for physics-based functionals.

C. Exchange-hole functional

Exchange is exactly a hole integral,

$$E_x[\rho] = \frac{1}{2} \int d^3r_0 \rho(r_0) \int d^3u \frac{n_x(r_0, \mathbf{u})}{u}, \quad (10)$$

where $\mathbf{u}$ is the inter-electron displacement and $u = |\mathbf{u}|$: around each electron at r_0 sits an exchange hole $n_x \leq 0$ that removes one electron from the surroundings ($\int n_x d^3u = -1$) and equals $-\frac{1}{2}\rho(r_0)$ on top, and E_x is the electron-hole electrostatic attraction. Because the $1/u$ weight retains only the spherically averaged ($\ell = 0$) hole, we expand that monopole *directly* in the scale-free SIMPLE radial basis, factoring out its on-top amplitude $-\frac{1}{2}\rho$ so that the coefficients are dimensionless,

$$\bar{n}_x(r_0, u) = -\frac{1}{2}\rho(r_0) \sum_n \tilde{\varrho}_n(r_0) R_n(u/R_{\text{ad}}), \quad (11)$$

so the hole is a dimensionless coefficient vector $\tilde{\varrho}$ — a SIMPLE coefficient like $\varrho_{n\ell m}$, with the prefactor restoring the density units, exactly as in Eq. (9) (the $-\frac{1}{2}\rho$ on-top amplitude here in place of the $A_0 \bar{\rho}_{\sigma, \text{safe}}$ amplitude

there) — evaluated at the dimensionless displacement $t = u/R_{\text{ad}}$ on the adaptive radius $R_{\text{ad}} = \xi^*/k_F$ [Eq. (2)]. The $1/u$ weight then collapses the energy per electron to a single linear contraction onto the basis Coulomb moments $\mathbf{C}$ of Eq. (9),

$$\varepsilon_x(r_0) = -\pi \rho(r_0) R_{\text{ad}}^2 \tilde{\varrho} \cdot \mathbf{C}, \quad C_n = \int_0^1 R_{n0}(t) t^2 dt, \quad (12)$$

the unit-frame Coulomb moments of the basis — the same $1/u$ weighting as Eq. (9), now on the adaptive unit frame. The two exact hole constraints are likewise linear contractions. The *sum rule*, that the hole removes exactly one electron ($\int n_x d^3u = -1$), reads $-2\pi \rho R_{\text{ad}}^3 \tilde{\varrho} \cdot \mathbf{B} = -1$ with the charge moments $B_n = \int_0^1 R_{n0}(t) t^2 dt$ (the $4\pi u^2 du$ volume weight). The *on-top* condition, that the hole equals $-\frac{1}{2}\rho$ at electron coincidence, fixes the shape normalization $\tilde{\varrho} \cdot \mathbf{R}_0 = 1$ [$R_{0,n} = R_{n0}(0)$]. Only the $\mathbf{C}$ -projection of the hole reaches the energy.

A valid exchange functional must satisfy: (C1) negativity and the sum rule; (C2) spin scaling; (C3) uniform-density scaling $E_x[\gamma^3 \rho(\gamma r)] = \gamma E_x[\rho]$; (C4) the homogeneous-gas (LDA) limit $F_x \rightarrow 1$; (C5) the Fermi-Amaldi one-electron limit, which existing semilocal functionals do not satisfy; (C6) the second-order gradient expansion $F_x \rightarrow 1 + \frac{10}{81} s^2$; and (C7) the Lieb-Oxford bound $F_x \leq 1.804$, where $F_x = \varepsilon_x/\varepsilon_x^{\text{unif}}$ is the enhancement factor and $\varepsilon_x^{\text{unif}} = -\frac{3}{4}(3/\pi)^{1/3} \rho^{1/3}$. We produce $\tilde{\varrho}$ by a parameter-free map that meets (C1)–(C6) by construction and respects (C7).

LDA and Fermi-Amaldi limits. At uniform density $\tilde{\varrho}$ is the homogeneous-gas hole $S(x) = [3j_1(x)/x]^2$ projected onto the basis. Because the practical window is finite ($R_c = 6$ bohr) the projection alone slightly misses $F_x = 1$; we therefore *moment-match* it to the hole's exact charge, on-top, and Coulomb moments, which restores $F_x = 1$ exactly despite the truncation (the matching is needed only at finite R_c — it becomes automatic as $R_c \rightarrow \infty$; App. F). This LDA hole $\tilde{\varrho}_{\text{LDA}}$ anchors the map. Where the window cannot enclose a spin pair — a one-electron core or density tail, per-spin charge $Q_\sigma < 1$ — no screened scale exists and the hole shape becomes the density-following $-\varrho_0/Q_\sigma$ (the negative neighboring density, with dimensionless monopole coefficients $\varrho_0 = \{\varrho_{n00}\}$ normalized to the per-spin enclosed charge Q_σ ; App. F), the exact one-electron self-interaction correction (C5). A smooth charge gate blends the two, $(1 - W_{\text{FA}}) \tilde{\varrho}_{\text{bulk}} + W_{\text{FA}} \tilde{\varrho}_{\text{FA}}$, with $W_{\text{FA}}(Q_\sigma) = 1$ for $Q_\sigma \leq 1$ (pure Fermi-Amaldi), 0 for $Q_\sigma \geq 2$ (the screened bulk hole), and the quintic smoothstep $W_{\text{FA}} = 1 - t^3(6t^2 - 15t + 10)$, $t = Q_\sigma - 1$, in between (C^2 at both joins); it reproduces H and spin-paired He to ~ 2 mHa. Forming $\tilde{\varrho}$ on the adaptive radius makes uniform scaling (C3) exact as $R_c \rightarrow \infty$ and close within the data window (Fig. 2).

Gradient expansion from the kernel. Inhomogeneity enters through every SIMPLE channel — the per- ℓ squared distances of the features from the uniform gas

— and the kernel uses all of them as coordinates. The second-order gradient expansion, however, constrains only the $\ell = 1$ channel, whose squared distance from the uniform gas is the reduced gradient s^2 [Eq. (8)]. We produce the bulk coefficients by interpolating the dimensionless hole shape $\tilde{\mathbf{q}}$ directly over fixed points with a kernel,

$$\tilde{\mathbf{q}}(\mathbf{x}) = \tilde{\mathbf{q}}_{\text{LDA}} + \sum_k \mathcal{K}(\mathbf{x}, \mathbf{x}_k) \mathbf{c}_k, \quad (13)$$

where $\mathbf{x}$ collects the per- ℓ SIMPLE distances ($\ell = 0$: the monopole power vector; $\ell = 1$: s^2) and the HEG node $\mathbf{x} = 0$ pins $\tilde{\mathbf{q}} = \tilde{\mathbf{q}}_{\text{LDA}}$. Because the energy is linear in $\tilde{\mathbf{q}}$ [Eq. (12)], the enhancement is a linear functional of a single moment of the shape deviation $\delta\tilde{\mathbf{q}} = \tilde{\mathbf{q}} - \tilde{\mathbf{q}}_{\text{LDA}}$,

$$F_x - 1 = \kappa \delta\tilde{\mathbf{q}} \cdot \mathbf{C}, \quad \kappa = \frac{\pi \xi^{*2}}{(3\pi^2)^{2/3} |\varepsilon_x^{\text{unif}} / \rho^{1/3}|}, \quad (14)$$

density-independent by the scale-free identity $R_{\text{ad}}^2 \rho^{2/3} = \xi^{*2} / (3\pi^2)^{2/3}$. The $\ell = 1$ channel of the kernel — a single node along the gradient axis — has its amplitude fixed so this slope reproduces the second-order gradient expansion exactly,

$$F_x \xrightarrow{s \rightarrow 0} 1 + \frac{10}{81} s^2, \quad (15)$$

with no separate gradient term and no fitted number; parity (F_x even in $\nabla\rho$) is automatic because the kernel coordinate is s^2 , and the fourth-order term is left unconstrained (App. G). Fixing the slope leaves a single open length scale — the width of the $\ell = 1$ kernel node — a free parameter held at a moderate value. The resulting slowly-varying enhancement peaks just above LDA and stays well below the Lieb–Oxford ceiling $F_x \leq 1.804$ (C7 [7, 8], respected not saturated; Fig. 1), the node amplitude holding the 10/81 slope. Along the $\ell = 0$ (power-spectrum) axis the kernel still interpolates a deviation that decays away from the HEG signature, so on real atoms — whose local power spectra depart from the homogeneous gas — the enhancement remains a *bounded deformation about* LDA. Reference-free this leaves the functional close to LDA on atoms; strongly inhomogeneous (open-shell) atoms are improved by adding exact atomic holes as further kernel fixed points, each carrying only the deviation beyond LDA and vanishing far from its reference, so the LDA, Fermi–Amaldi, and gradient limits are untouched (App. G).

Spin. The exchange energy decomposes exactly as $E_x[\rho_\uparrow, \rho_\downarrow] = \frac{1}{2}(E_x[2\rho_\uparrow] + E_x[2\rho_\downarrow])$, so the spin-paired functional follows from the total-density expressions with the on-top count of 2, and the open-shell case from applying the spin-scaling relation to each channel (App. H). Constraints (C1)–(C6) are thus satisfied by construction rather than by fitting, and the Lieb–Oxford bound (C7) is respected (the enhancement stays below 1.804; Fig. 1); the one remaining length scale, the $\ell = 1$ kernel width, is a free parameter. Those constraints requiring the kinetic-energy density — the single-orbital Lieb–Oxford bound

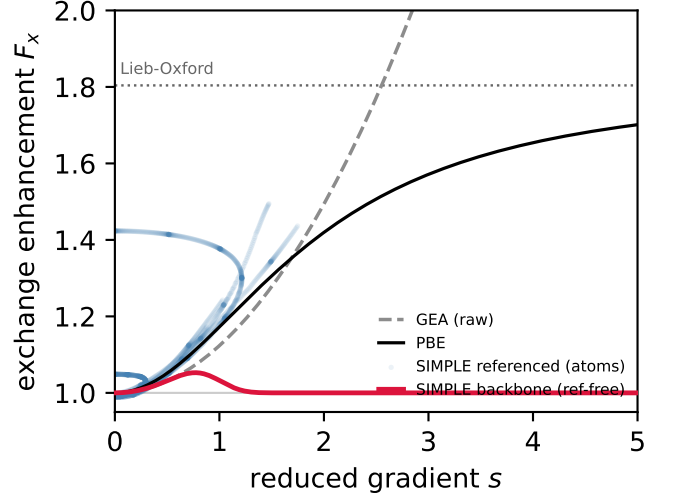


FIG. 1. Exchange enhancement $F_x = \varepsilon_x / \varepsilon_x^{\text{unif}}$ of the SIMPLE kernel hole. *Crimson*: the reference-free backbone on the homogeneous gas at reduced gradient s (the pure-GEA-axis feature point). The single $\ell = 1$ kernel node fixes the small- s slope to the exact second-order value $\mu = 10/81$ [Eq. (15)]; the bounded deformation peaks just above LDA (≈ 1.05) and relaxes back, so reference-free the hole is essentially LDA+Fermi–Amaldi. *Blue*: the realized enhancement of the *final referenced* functional evaluated pointwise on real closed-shell atoms (Ne, Ar, Kr; energy-relevant region $\rho > 0.05$). The exact-hole references lift the enhancement from the reference-free ≈ 1.05 to PBE-like values (≈ 1.4) at the gradients atoms actually carry — the origin of the atomic binding (Table II). Raw pointwise GEA (grey dashed) diverges and PBE [8] (black) saturates; the dotted line is the Lieb–Oxford ceiling $F_x = 1.804$. The on-top value is preserved for any amplitude, so the Fermi–Amaldi limit is untouched.

$F_x(s, \alpha=0) \leq 1.174$ and one-electron self-correlation freedom — lie outside this orbital-free, exchange-only construction. The hole is spherically symmetric and uses no multipoles beyond the monopole; carrying explicit $\ell \geq 1$ hole channels is the natural route to a richer functional (App. G).

D. Numerical implementation

Convolutions and spectral operators. Every descriptor is a fixed-stencil convolution of the density. The reduced-gradient ingredient of Sec. IB 1 is evaluated by a parameter-free spectral operator: the $\ell = 1$ slope-at-origin sum $\rho'(r_0) = \sum_n a_n^{(1)} R'_{n1}(0)$ (with a constant-annihilation correction enforcing the uniform-density limit; $a_n^{(1)}$ the $\ell = 1$ channel coefficients), a fixed linear stencil on the gridded density. So formed, $|\nabla\rho|$ is recovered to $\sim 1\%$ in the core/valence of smooth (pseudopotential) atoms (App. E).

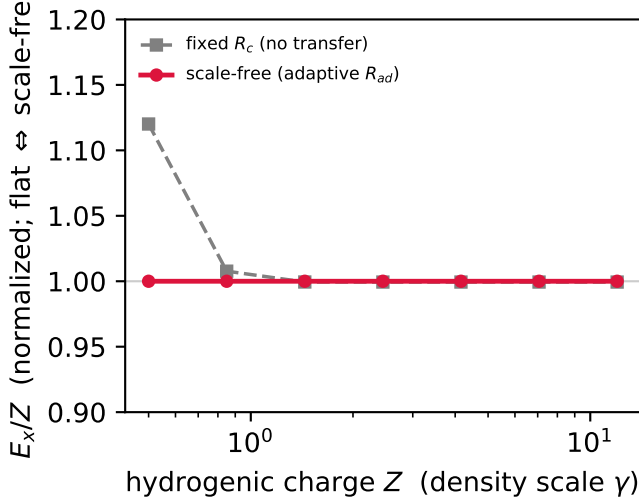


FIG. 2. Uniform-scaling (scale invariance) of the hole on atoms. For the hydrogenic density $\rho_Z = (Z^3/\pi)e^{-2Zr}$ the map $Z \rightarrow \gamma Z$ is the uniform scaling $\rho \rightarrow \gamma^3 \rho(\gamma r)$, so the exact constraint $E_x[\rho_\gamma] = \gamma E_x[\rho]$ requires E_x/Z to be Z -independent. The scale-free hole (adaptive radius R_{ad} , reached by the transfer) is flat to machine precision; the fixed- R_c hole (no transfer) drifts once the density is diffuse enough to overflow the window. R_c remains the explicit data window, and scale invariance holds within it.

Self-consistent potential. The exchange potential is the explicit variational derivative $v_x = \delta E_x / \delta \rho$, which for convolutional descriptors is itself a convolution over the kernel [Eq. (I2)], following Ref. [6]; in the radial implementation it is the transpose of the forward stencils. The on-top scale $\zeta(r_0)$ is obtained without an inner self-consistency loop: $Q_S(\zeta) = \mathbf{B}(\zeta) \cdot \mathbf{g}_0$ is monotonic in ζ , so inverting $Q_S = 2$ pointwise on the precomputed envelope table is the same fixed-stencil enclosed-charge solve used for the adaptive radius R_{ad} (Sec. IA 1), vectorized over all grid points (App. I). The asymptotic $-1/r$ exchange tail is recovered by an energy-neutral gauge fix plus a low-density damping (App. I). The $\ell = 1$ gradient channel [Eq. (15)] enters the potential through its own adjoint, which — growing only as k_n^1 — is well-conditioned, so the second-order functional is self-consistent and stable without any auxiliary loop (App. I).

Code and tests. The functional is implemented in the `atom-dft` finite-element radial Kohn–Sham code [9, 10] (all-electron and pseudopotential). Descriptor tests use two independent grids — a radial finite-element grid and a 3D Cartesian finite-difference grid carrying pseudo- H_2/N_2 densities. The test harness was driven in part by an agentic AI assistant, with every reported result reproduced by the shipped scripts.

II. RESULTS

A. Numerical validation of SIMPLE feature properties

The features are validated on two independent grids — a radial finite-element grid and genuine 3D Cartesian lattices carrying superposition densities (pseudo- H_2 , pseudo- N_2 , pseudo- H_2O , and FCC bulk Al and Pt) — at realistic settings, to establish the practical limits of each feature family. Unless noted, $\ell_{\max} = 3$.

1. Orthogonality and limiting behavior

The features reproduce the analytic HEG values $\varrho_{n00}^{\text{HEG}} = (-1)^n / (n+1)$, $\varrho_{n\ell m} = 0$ ($\ell > 0$) to $\sim 2 \times 10^{-11}$ in the continuum and to $< 4 \times 10^{-3}$ on the $h = 0.2$ bohr grid (SI Fig. 6), degrading only as the channel count approaches the Nyquist bound $n_{\text{in}} \lesssim R_c/2h$. As the density is scaled toward vacuum the coefficient norm and power spectrum decay as $\mathcal{O}(\epsilon)/\mathcal{O}(\epsilon^2)$ (fitted slope 2.000; SI Fig. 7), so the features vanish smoothly rather than diverging — the property that lets them be evaluated in the tail regions of real atoms.

2. Radial scale and scale invariance

Scale invariance. Under $\rho \mapsto \lambda^3 \rho(\lambda r)$ the relative deviation at $\lambda = 2$ is 0.35–1.7% in the continuum, versus 170% for the untransformed fixed-window features (Fig. 3); the residual is set by the inner-channel resolution bound and the low-density window clamp of Sec. IA 2.

Choosing R_c and n_{out} by an energy scale. The window cutoff and channel count are fixed not by inspection but by a target energy accuracy. Evaluating the Hartree self-energy of the hydrogen 1s density through the SIMPLE Coulomb projection [Eq. (9)] against the exact 5/16 Ha (Fig. 4), the window (Coulomb-tail) floor reaches the milli-Hartree scale only at $R_c \gtrsim 6$ bohr (−1.7 mHa at $R_c = 5$, −0.4 mHa at $R_c = 6$), while the radial-basis error falls below 1 mHa by $n_{\text{out}} \approx 8$ and reaches that floor by $n_{\text{out}} = 10$ (−0.5 mHa). On a real Cartesian grid this tolerance is registration-robust for $h \lesssim 0.2$ –0.25 bohr and reaches the ± 1 mHa edge only at $h = 0.3$ (Fig. 4c). We therefore adopt $R_c = 6$ bohr and $n_{\text{out}} = 10$ throughout.

Cost. With $n_{\text{in}} = R_c/h$ grid-resolved inner channels the scale transform is invariant up to a dilation $\Lambda_{\max} \simeq n_{\text{in}}/n_{\text{out}}$, and the feature set costs $N_{\text{conv}} = n_{\text{in}}(\ell_{\max} + 1)^2$ convolutions. The cost is set by the target scale range, not the grid: fixing $n_{\text{in}} = n_{\text{out}}\Lambda$ holds N_{conv} fixed across h (finer grids only sharpen the same channels; App. L). At $n_{\text{in}} = 20$ (covering $\Lambda = 2$ for $n_{\text{out}} = 10$) the full feature set is $N_{\text{conv}} = 320, 180, 80$ convolutions for $\ell_{\max} = 3, 2, 1$ on any grid $h \leq 0.3$ bohr.

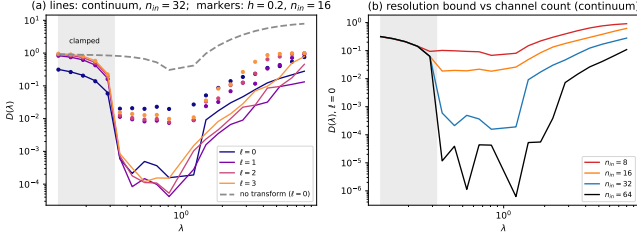


FIG. 3. Scale invariance of the SIMPLE features under uniform scaling $\rho \mapsto \lambda^3 \rho(\lambda \mathbf{r})$ (Sec. I A 2) and its two breakdown mechanisms; $D(\lambda) = \|\varrho(\lambda) - \varrho(1)\|/\|\varrho(1)\|$ is the relative deviation of the feature vector from its unscaled ($\lambda = 1$) value. (a) $D(\lambda)$ for the channels $\ell = 0$ –3: solid lines are the continuum quadrature ($n_{\text{in}} = 32$ inner channels) and markers the $h = 0.2$ bohr grid with the grid-matched count ($n_{\text{in}} = 16$). The gray dashed line is the untransformed fixed-window baseline ($\ell = 0$, no scale transform), which deviates at the $\mathcal{O}(1)$ ($\sim 170\%$) level. In the shaded region ($\lambda < \lambda_{\text{clamp}} \approx 0.32$) the adaptive radius is clamped at the window ($R_{\text{ad}} = R_c$) and the features revert continuously to fixed-window behavior. (b) The resolution mechanism isolated ($\ell = 0$, continuum): $D(\lambda)$ for increasing inner-channel counts $n_{\text{in}} \in \{8, 16, 32, 64\}$; the deviation falls systematically with n_{in} , and the usable dilation range grows as $k^* \lambda \lesssim n_{\text{in}} \pi / R_c$ (on a grid $n_{\text{in}} \lesssim R_c / 2h$).

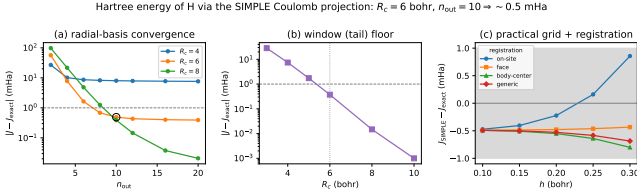


FIG. 4. Calibrating R_c and n_{out} by an energy scale: the Hartree self-energy of the hydrogen $1s$ density through the SIMPLE Coulomb projection [Eq. (9)] versus the exact $5/16$ Ha. (a) radial-basis convergence vs n_{out} for several windows (circle: the adopted $R_c = 6$, $n_{\text{out}} = 10$ at -0.5 mHa); a short window ($R_c = 4$) is Coulomb-tail-limited and never reaches 1 mHa. (b) the n_{out} -converged window floor vs R_c , crossing 1 mHa at $R_c \approx 6$ bohr. (c) the same energy on a 3D Cartesian grid vs spacing h and sub-grid registration: registration-robust within the ± 1 mHa band (shaded) for $h \lesssim 0.25$ bohr.

3. Stability of invariants on Cartesian grids

We assess the invariants at the fixed production settings ($R_c = 6$ bohr, $n_{\text{out}} = 10$, $n_{\text{in}} = 20$, so $\Lambda_{\text{max}} = n_{\text{in}}/n_{\text{out}} = 2$), pooled over the five Cartesian systems (Fig. 5; full breakdown in App. K). The power spectrum, bispectrum, and reduced gradient s are robust: random rotation and sub-grid translation move them by $\lesssim 1\%$ (a few percent only at the bare pseudo- H_2 cusp), grid refinement converges as $\mathcal{O}(h^2)$ ($\leq 3\%$ at $h = 0.2$), and uniform scaling within Λ_{max} holds the power spectrum and bispectrum to $\sim 1\%$ and the reduced gradient to $\lesssim 10\%$ (median $\sim 7\%$; the gradient is the most

resolution-demanding feature and tightens further with n_{in} , App. K). The reduced Laplacian q is the exception on two counts: where its magnitude is small it is rotation/translation-noisy, and — decisively — its reconstruction against the exact $\nabla^2 \rho$ is wrong by factors of tens to hundreds in dense bulk environments (Fig. 5b). q is therefore excluded.

The two panels of Fig. 5 measure the same finite-resolution reconstruction error at different points: the true reduced gradient is exactly scale-invariant, so panel (b) reports the reconstruction error at the unscaled density ($\sim$ few percent at $n_{\text{in}} = 20$), while the scaling column of panel (a) reports how that error *changes* with λ — it grows as the dilated density shifts content to higher wavevectors that the fixed n_{in} resolves less well, reaching $\sim 10\%$ by $\lambda \sim 1.5$. Both shrink together as n_{in} increases; neither reflects any motion of the exact s .

Reduced gradient in energies. The reconstructed s tracks a direct finite-difference gradient to within ~ 5 –6% at the atom-site and off-axis centers of pseudo- H_2 and pseudo- N_2 , and — more importantly — carries the energy faithfully: feeding the SIMPLE s into the PBE exchange enhancement and evaluating the exchange energy non-self-consistently on fixed atomic densities (H, N valence) reproduces PBE-on-finite-difference- s to within $\sim 4\%$ of the LDA $\rightarrow$ PBE gradient correction, i.e. under 0.4% of E_x (script `cartesian_s_pbe.py`). The SIMPLE-derived gradient is therefore practically reliable at the default $R_c = 6$, $n_{\text{out}} = 10$.

Practical ceiling. Taken together, the usable regime for the features is $h \lesssim 0.2$ –0.25 bohr (1 mHa Coulomb accuracy, registration-robust), $\ell_{\text{max}} \leq 3$ (cost $\propto (\ell_{\text{max}} + 1)^2$), and the second-order gradient s as the functional ingredient, with the reduced Laplacian q excluded.

B. Exchange-hole functional

Being spherically symmetric, the functional uses only the $N_C = 20$ monopole channels (plus the fixed $\ell = 1$ gradient operator), so a self-consistent step costs $2N_C$ convolutions — the forward monopole coefficients $\{\varrho_{n00}\}$ and the adjoint potential [Eq. (12)] — on a single cutoff $R_c = 6$ bohr. The map carries no fitted parameters: the kernel is anchored at the homogeneous-gas node (LDA) and its single $\ell = 1$ node amplitude is fixed by the gradient expansion [Eq. (15)], with the node width a single free length scale (the Lieb–Oxford bound, C7, is respected, not saturated). We report the construction in three stages: its exact limits and stability, the reference-free baseline, and the route to atomic binding.

Exact limits and stability. The realized functional meets its design limits to the basis tolerance. The homogeneous-gas enhancement is $F_x = 1$ to machine precision (C4); the single $\ell = 1$ kernel node reproduces the second-order gradient slope $\mu = 10/81$ to $\sim 2\%$ (C6, Fig. 1); and the one-electron Fermi–Amaldi channel reproduces H and spin-paired He to ~ 2 mHa (C5).

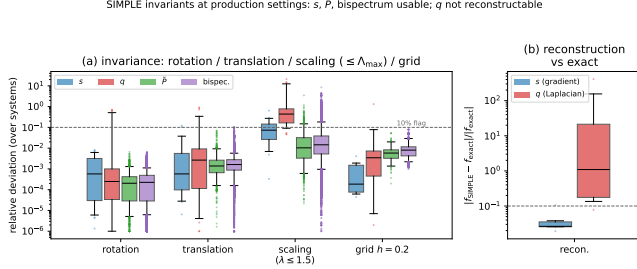


FIG. 5. SIMPLE invariants at the fixed production settings ($R_c = 6$ bohr, $n_{\text{out}} = 10$, $n_{\text{in}} = 20$; $\Lambda_{\text{max}} = n_{\text{in}}/n_{\text{out}} = 2$), pooled over the five test systems (box: quartiles; whiskers: 5–95%; points: outliers; dashed line: the 10% flag). **(a) Invariance** — relative deviation of each family (s , q , power spectrum $\tilde{P}$, bispectrum) under random rotation, sub-grid translation, uniform scaling (within $\lambda \leq \Lambda_{\text{max}}$), and grid refinement vs the continuum. Rotation, translation, and grid refinement are well controlled; under scaling the power spectrum and bispectrum stay near 1% and the reduced gradient near 10% (it is the most resolution-demanding and tightens with n_{in} ; App. K). **(b) Reconstruction accuracy** — relative error of the reconstructed s and q against the exact (finite-difference) gradient/Laplacian: s is accurate ($\sim$ few percent), whereas q is wrong by factors up to $\sim 10^2$ in dense environments. The reduced gradient and the spectra are thus both invariant and accurate, while the reduced Laplacian is not reconstructable and is excluded. Detailed per-test panels are in App. K.

The discrete adjoint potential [Eq. (12)] matches a finite-difference derivative of the energy to a relative error $\lesssim 10^{-4}$, and the functional is self-consistently stable: SCF converges for every atom tested.

Reference-free baseline. With no atomic references the kernel reduces, by construction, to the LDA hole plus the Fermi–Amaldi correction. The gradient enhancement is a bounded deformation confined to the slowly-varying region (Fig. 1) and does not reach the strong inhomogeneity of an atomic core, so the self-consistent energies track LDA+FA (Table I): He is near-exact (the Fermi–Amaldi limit, +3 mHa), while the denser closed shells underbind — Be +0.34, Ne +0.51, Mg +0.55 Ha versus exact exchange. This is the honest cost of a reference-free hole: the second-order gradient term is correct but, being a small slowly-varying-limit correction, is not by itself sufficient to bind an atom, and the reference-free functional is less accurate than PBE or rSCAN.

Atomic references: the referenced functional. Binding comes from adding exact atomic exchange holes as further kernel fixed points (Sec. IC). Each reference carries only the deviation beyond LDA and decays away from its own feature signature, so it sharpens the strongly inhomogeneous region while leaving the LDA, Fermi–Amaldi, and gradient limits intact. Populating the kernel with exact holes that cover the closed-shell density manifold — sampled wherever the windowed hole is well defined, $\rho \gtrsim 10^{-2}$ — yields the final functional. Evaluated on the exact density, it reproduces closed-shell atomic exchange

TABLE I. Exchange-only energies (Ha) for closed-shell atoms: the reference-free SIMPLE kernel hole (this work, no atomic references) versus exact exchange (HF), with PBE [8] and rSCAN [11] for reference. Reference-free the functional reduces to LDA+Fermi–Amaldi — near-exact for He but under-binding the denser shells; atomic binding requires the exact-hole references (text). Spin treatment per Sec. IC; open-shell N, P require the unrestricted reference and are deferred.

Atom	exact-x	SIMPLE (ref-free)	PBE	rSCAN
He	−1.0019	−1.0054	−0.9705	−0.9955
Be	−1.9215	−1.5827	−1.8853	−1.9105
Ne	−5.4012	−4.8918	−5.2989	−5.4110
Mg	−6.8576	−6.3111	−6.7599	−6.8878

TABLE II. Closed-shell atomic exchange errors (mHa) of the *referenced* SIMPLE kernel hole, evaluated on the exact density, versus exact exchange, with PBE [8] and rSCAN [11]. The kernel is populated with exact holes covering the closed-shell density manifold ($\rho \gtrsim 10^{-2}$). SIMPLE’s mean absolute error (34 mHa) is below PBE (45); rSCAN (14) additionally uses the kinetic-energy density. The residual concentrates in the alkaline-earth ns^2 shells (Fermi–Amaldi-dominated).

Atom	SIMPLE	PBE	rSCAN
He	−6	23	1
Be	99	44	17
Ne	11	78	2
Mg	−58	105	6
Ar	10	62	−5
Ca	−70	73	10
Zn	−14	19	30
Kr	6	35	−26
Sr	−73	57	6
Pd	4	−3	19
Cd	−21	22	28
Xe	−8	24	−30
Ba	−86	35	−8
Hg	−13	−45	10
MAE	34	45	14

to a mean absolute error of 34 mHa across the fourteen closed-shell atoms from He to Hg, *below* PBE (45 mHa) and approaching rSCAN (14 mHa, which additionally consumes the kinetic-energy density) (Table II). The noble-gas and filled- d atoms are reproduced to ~ 10 mHa (Ne 11, Ar 10, Kr 6, Pd 4); the residual error concentrates in the alkaline earths (Be, Ca, Sr, Ba), the low-enclosed-charge ns^2 shells where the Fermi–Amaldi channel dominates the hole. The reference coverage of the density manifold, rather than the descriptor resolution, sets this accuracy. This is the exchange-energy accuracy on the exact density; the self-consistent referenced functional — which additionally requires conditioning the low-density reference block — and the open-shell atoms (which require the unrestricted exact-exchange reference) are the next stage.

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Appendix A: The orthonormal local expansion

We expand the density inside a ball B_{R_c} in a product basis of real spherical harmonics and radial functions. The radial functions are spherical Bessel functions truncated to the ball with Dirichlet (vanishing) boundary conditions,

$$R_{n\ell}(r) = \mathcal{N}_{n\ell} j_\ell\left(\frac{z_{n\ell} r}{R_c}\right), \quad j_\ell(z_{n\ell}) = 0, \quad (\text{A1})$$

where $z_{n\ell}$ is the $(n+1)$ th positive zero of j_ℓ . The constants $\mathcal{N}_{n\ell}$ are fixed by orthonormality with the radial measure $r^2 dr$, $\int_0^{R_c} r^2 R_{n\ell} R_{n'\ell} dr = \delta_{nn'}$, which follows from the Sturm–Liouville orthogonality of the Bessel functions. Why this basis? The j_ℓ are eigenfunctions of the radial Laplacian, $\nabla^2[j_\ell(kr)Y_{\ell m}^{\mathbb{R}}] = -k^2 j_\ell(kr)Y_{\ell m}^{\mathbb{R}}$,

so derivatives act diagonally; and the Dirichlet truncation makes the set complete and orthonormal on the finite ball, which is what lets integrals collapse to sums (App. D).

The coefficients $c_{n\ell m}(\mathbf{r}_0) = \int_{B_{R_c}} K_{n\ell m}(\mathbf{r}) \rho_\sigma(\mathbf{r}_0 + \mathbf{r}) d^3r$ are an exact, invertible representation of the local density up to the truncation $n < n_{\text{in}}$, $\ell \leq \ell_{\text{max}}$: by orthonormality $\rho_\sigma(\mathbf{r}_0 + \mathbf{r}) = \sum_{n\ell m} c_{n\ell m}(\mathbf{r}_0) K_{n\ell m}(\mathbf{r})$ inside the ball. Each coefficient is a convolution of the density with a fixed kernel, hence a single fixed stencil per channel on a grid.

A useful identity, used repeatedly, is the monopole integral of a basis function,

$$A_n(R) = \int_{B_R} K_{n00} d^3r = \sqrt{4\pi} \int_0^R r^2 R_{n0}(r) dr, \quad (\text{A2})$$

which for the Dirichlet basis evaluates in closed form and gives the constant-density projection $c_{n00} = \rho A_n(R)$, with the ratio $A_n/A_0 = (-1)^n/(n+1)$ used below.

Appendix B: The scale transform

This appendix derives the scale-invariance construct of Sec. IA 2.

The problem. We want features that do not change when the density is uniformly scaled, $\rho_\lambda(\mathbf{r}) = \lambda^3 \rho(\lambda \mathbf{r})$ (this normalization keeps the electron count fixed). Exact scale invariance is obtained by choosing the cutoff to follow the local Fermi wavelength, $R_c \propto k_F^{-1}$; but a density-dependent cutoff cannot be implemented as a single fixed convolution stencil. SIMPLE keeps a fixed physical window and *reconstructs* the adaptive-window descriptors by a linear map.

The dilation identity [Eq. (3)]. Substitute $\mathbf{r} \rightarrow \mathbf{r}/\lambda$ in the projection Eq. (1) and evaluate the coefficients of the scaled density at the scaled cutoff R/λ . Because each basis function depends on r only through the dimensionless ratio $z_{n\ell} r/R$, the integrand keeps its shape and only prefactors survive: the λ^3 from ρ_λ , a λ^{-3} from d^3r , and a $\lambda^{-3/2}$ from the normalization $\mathcal{N}_{n\ell} \propto R^{-3/2}$ combine to the $\lambda^{3/2}$ of Eq. (3). At $\ell = 0$ this says the fixed-window coefficients are samples of the spherical Bessel transform of the unscaled density at *dilated* wavevectors: scaling moves the sampling points but loses no information, provided the window still contains the density.

Why subtract the window mean [Eqs. (4)–(5)]. The Dirichlet basis vanishes at R_c , but a generic density does not, so directly re-projecting carries slowly-decaying boundary oscillations (for the HEG, where the density is pure boundary mismatch, this is a 10% effect). Transferring the fluctuation about the window average and adding back the mean analytically through the monopole integrals A_n [Eq. (A2)] removes the effect: for a uniform

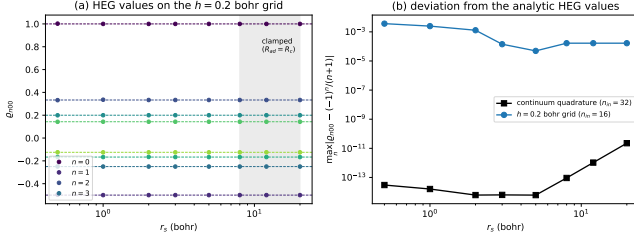


FIG. 6. Numerical verification of the homogeneous-electron-gas (HEG) limit through the full SIMPLE pipeline (window projection, adaptive radius, mean-split transfer, non-dimensionalization), referenced from Sec. II A. (a) The monopole descriptors q_{n00} on the $h = 0.2$ bohr grid (points) against the analytic HEG values $q_{n00}^{\text{HEG}} = (-1)^n/(n+1)$ (dashed), over $r_s \in [0.5, 20]$; the shaded region marks densities where the adaptive radius is clamped ($R_{\text{ad}} = R_c$), which leaves the HEG values unaffected. (b) Maximum deviation from the analytic values versus r_s : the continuum pipeline reproduces them to $\sim 2 \times 10^{-11}$ (exact by construction; residual from the root-find tolerance), and the $h = 0.2$ bohr grid to $< 4 \times 10^{-3}$ (kernel-quadrature error).

density the fluctuation $\tilde{c} \equiv 0$, so the transfer is exact at any truncation — this is why the HEG limit is exact by construction (Fig. 6).

The closed form [Eq. (6)]. The transfer matrix is the radial overlap of an adaptive-window basis function with a fixed-window one. For two solutions $j_\ell(ar), j_\ell(br)$ of the spherical Bessel equation ($a = z_{m\ell}/R_{\text{ad}}, b = z_{n\ell}/R_c$) the Sturm–Liouville cross-product identity integrates the overlap in closed form, Eq. (6), with no quadrature or tabulation. At $R = R_{\text{ad}}$ the first bracket term vanishes because $j_\ell(aR_{\text{ad}}) = j_\ell(z_{m\ell}) = 0$. (Tabulating in the ratio R_{ad}/R_c fails: the high- n elements oscillate on the scale $\pi/z_{n\ell}$, so naive interpolation is inaccurate — the closed form is load-bearing.) Near-degenerate pairs $a \approx b$ use the analytic limit of Eq. (6); when $R_{\text{ad}} = R_c$ (clamped) the transfer is the identity. Crucially $M^{(\ell)}$ acts only on the radial index n , identically for every m — the property that makes the transform commute with rotations (App. C).

Appendix C: Non-dimensionalization and invariance

The descriptors are normalized at the adaptive radius, $q_{n\ell m} = c'_{n\ell m}/[A_0(R_{\text{ad}})\bar{\rho}_{\sigma, \text{safe}}(R_{\text{ad}})]$. Under scaling, $A_0(R_{\text{ad}}/\lambda) = \lambda^{-3/2}A_0(R_{\text{ad}})$ and $\bar{\rho}_\sigma \mapsto \lambda^3\bar{\rho}_\sigma$, so the denominator scales as $\lambda^{3/2}$, exactly cancelling the $\lambda^{3/2}$ of the numerator from Eq. (3): $q_{n\ell m}$ is invariant for every ℓ , up to ρ_{min} effects and the transfer truncation. (This corrects an earlier convention that multiplied by $\bar{k}_F^\ell$, which actually multiplies the fixed- ξ features by λ^ℓ and breaks the invariance it was meant to support; an order-unity balance across ℓ may instead be restored by a *constant* $(\xi^*)^{-\ell}$, which affects neither covariance nor invariance.)

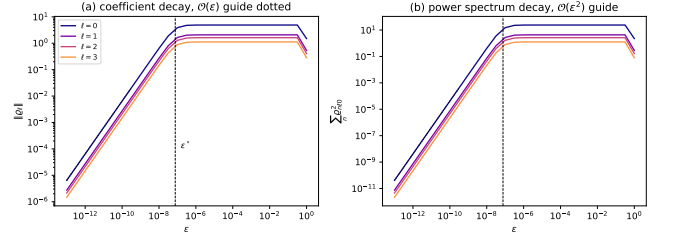


FIG. 7. Low-density (vacuum) stability of the descriptors as the density is scaled down uniformly, $\rho \mapsto \epsilon\rho$, over thirteen decades of ϵ . The descriptor-coefficient norm $\|q_{n\ell m}\|$ (blue) decays as $\mathcal{O}(\epsilon)$ and the power spectrum $\tilde{P}_{n\ell} = \sum_m q_{n\ell m}^2$ (orange) as $\mathcal{O}(\epsilon^2)$ (reference slopes shown dashed; the fitted power-spectrum slope is 2.000), exactly as predicted in this section: below the crossover $\epsilon^* = \rho_{\text{min}}/\bar{\rho}_\sigma$ the safe floor $\bar{\rho}_{\sigma, \text{safe}}$ takes over and the descriptors vanish smoothly and without overflow rather than diverging by division by a vanishing mean density. This is the property that lets the functional be evaluated in near-vacuum tail regions of real atoms.

Rotational covariance. Under a rotation $\mathcal{R}$, the raw coefficients transform as rank- ℓ tensors, $c_{n\ell m} \mapsto \sum_{m'} D_{mm'}^{(\ell)}(\mathcal{R})c_{n\ell m'}$. The transform preserves this because (i) R_{ad} depends only on the invariant ball average; (ii) $M^{(\ell)}$ acts on n , not m ; (iii) the mean split touches only the invariant (0,0) channel. Since the transfer acts on the radial slot and the rotation on the magnetic slot, they commute,

$$(M^{(\ell)} \otimes \mathcal{K})(\mathcal{K} \otimes D^{(\ell)}) = (\mathcal{K} \otimes D^{(\ell)})(M^{(\ell)} \otimes \mathcal{K}), \quad (\text{C1})$$

so $q_{n\ell m}$ is a rank- ℓ tensor and any contraction to total angular momentum zero is invariant (verified numerically to $\lesssim 2 \times 10^{-15}$).

Low-density limit. With the safe floor $\bar{\rho}_{\sigma, \text{safe}} = (\bar{\rho}_\sigma^2 + \rho_{\text{min}}^2)^{1/2}$, as $\rho \rightarrow \epsilon\rho$ the adaptive radius clamps at the window and the descriptors reduce to fixed-window coefficients divided by a floor: $q_{n\ell m} = \mathcal{O}(\epsilon)$ and $\tilde{P}_{n\ell} = \sum_m q_{n\ell m}^2 = \mathcal{O}(\epsilon^2)$, smoothly and without overflow, with crossover at $\epsilon^* = \rho_{\text{min}}/\bar{\rho}_\sigma$.

Appendix D: Operator projections

Multilinear functionals as sums. Any ν -point integral functional with a rotation-invariant kernel, $\mathcal{O}_\nu[\rho](\mathbf{r}_0) = \int_{B_{R_c}} \cdots \int \mathcal{K}_\nu(\mathbf{r}_1, \dots, \mathbf{r}_\nu) \prod_i \rho(\mathbf{r}_0 + \mathbf{r}_i) d^3r_i$, becomes, on inserting the expansion $\rho(\mathbf{r}_0 + \mathbf{r}) = \sum c_{n\ell m} K_{n\ell m}(\mathbf{r})$,

$$\begin{aligned} \mathcal{O}_\nu[\rho] &= \sum_{\{n_i \ell_i m_i\}} T_{\{n_i \ell_i m_i\}}^{(\nu)} \prod_{i=1}^{\nu} c_{n_i \ell_i m_i}, \\ T^{(\nu)} &= \int \cdots \int \mathcal{K}_\nu \prod_i K_{n_i \ell_i m_i} d^3r_i. \end{aligned} \quad (\text{D1})$$

The projected-kernel tensor $T^{(\nu)}$ is computed once. Rotation invariance of $\mathcal{K}_\nu$ forces the magnetic indices to

couple to total angular momentum zero, which is exactly Clebsch–Gordan coupling; the power spectrum ($\nu = 2$), bispectrum ($\nu = 3$), and higher CG trees therefore form a complete basis for rotationally invariant multilinear functionals. Non-dimensionalization only rescales $T^{(\nu)}$ by scalar prefactors and does not change the admissible class.

The Coulomb operator is a monopole. For the exchange-relevant kernel $\mathcal{K} = 1/u$ acting on the density at displacement $\mathbf{u}$, the angular integral $\oint d\Omega_u (1/u) \rho(\mathbf{r}_0 + \mathbf{u})$ projects out only the $\ell = 0$ component, because $1/u$ is isotropic and the higher harmonics integrate to zero. Hence the projected Coulomb operator acts through the monopole descriptors alone,

$$\begin{aligned} \bar{\rho}(r_0, u) &= \frac{1}{4\pi} \oint \rho(r_0 + \mathbf{u}) d\Omega_u \\ &= A_0 \bar{\rho}_{\sigma, \text{safe}} \sum_n \varrho_{n00}(r_0) R_n(u), \end{aligned} \quad (\text{D2})$$

with the dimensionless monopole coefficients $\varrho_{n00} = (P_n \rho) / (A_0 \bar{\rho}_{\sigma, \text{safe}})$ [Eq. (7)], $\{P_n\}$ fixed window operators, and the local amplitude restoring the density units. The Coulomb operator acting on this monopole density is then a weighted sum [Eq. (9)],

$$\begin{aligned} \int_{B_{R_c}} \frac{\rho(r_0 + \mathbf{u})}{u} d^3u &= 4\pi \int_0^{R_c} \bar{\rho}(r_0, u) u du \\ &= A_0 \bar{\rho}_{\sigma, \text{safe}} \sum_n C_n \varrho_{n00}(r_0), \end{aligned} \quad (\text{D3})$$

with the basis Coulomb moments $C_n = 4\pi \int_0^{R_c} R_n(u) u du$. The $1/u$ weight reduces the radial measure from $u^2 du$ to $u du$: this is how the long-range $1/r$ structure that semilocal functionals miss enters, and it is the foundation of the exchange-hole functional below.

Appendix E: Reduced gradient from the expansion

The gradient expansion of exchange, $E_x = \int \varepsilon_x^{\text{LDA}}(\rho) [1 + \mu s^2 + \dots] \rho d^3r$, requires the reduced gradient $s = |\nabla \rho| / (2k_F \rho)$, with $k_F = (3\pi^2 \rho)^{1/3}$. We extract it from the SIMPLE expansion with a parameter-free spectral operator.

Gradient (slope at origin). The density at the expansion center along a ray is built from the $\ell = 1$ channel; its slope at r_0 is $\rho'(r_0) = \sum_n a_n^{(1)} R'_{n1}(0)$, where $a_n^{(1)}$ are the channel-1 radial expansion coefficients and $R'_{n1}(0) = \mathcal{N}_{n1} k_n / 3$ (the small-argument behavior $j_1(x) \approx x/3$), $k_n = z_{n1} / R_c$. This is a fixed linear operator on the gridded density, and — growing only as k_n^1 — it is well-conditioned, so the slope is recovered directly without any per-channel reformation.

Why not the Laplacian. The same construction would give the reduced Laplacian from the $\ell = 0$ channel via

$\nabla^2[R_{n0}] = -k_n^2 R_{n0}$, i.e. $\nabla^2 \rho(r_0) = -\sum_n a_n^{(0)} k_n^2 R_{n0}(0)$. The forward sum can be stabilized by forming it *per channel* (the $a_n^{(0)}$ decay for a smooth density, the growing k_n^2 weights applied afterward), but its variational adjoint — needed for the self-consistent potential — re-forms the single k_n^2 kernel $\sum_n (-k_n^2 R_{n0}(0)) R_{n0}(u)$, a near-singular ($\sim \nabla^2 \delta$) object whose huge oscillatory values are catastrophically ill-conditioned. The k_n^2 growth makes this fatal for $\ell = 0$, and no practical regularization kept the self-consistent cycle stable across cutoffs. We therefore stop the expansion at second order and do not carry a Laplacian channel; the functional uses only the $\ell = 1$ gradient.

Constant annihilation and the uniform limit. The $\ell = 1$ gradient operator ends with $\mathbf{O} \rightarrow \mathbf{O} - \text{diag}(\sum_j \mathbf{O}_{ij})$, forcing $\mathbf{O}[\text{const}] = 0$ (the gradient of a uniform density vanishes; the subtraction also removes the DC truncation residual), so the reduced gradient vanishes for the HEG by construction.

In SIMPLE features. Dividing this dimensional operator by the non-dimensionalization amplitude turns it into the feature contraction of Eq. (8): the spectral weights are $g_n^{(1)} \propto R'_{n1}(0)$ (slope-at-origin) acting on the $\ell = 1$ features ϱ_{n1m} . The reduced gradient is the norm of the $\ell = 1$ contraction — hence the $\ell = 1$ power of the spectrally-weighted features.

The 10/81 coefficient. For the ball-normalized reduced gradient the second-order gradient-expansion coefficient comes out to the standard $\mu = 10/81$ [12] on the single-scale path that atoms occupy, with $\kappa = s_{\text{SIMPLE}} / s_{\text{phys}} \rightarrow 1$ in the high-density limit — derived, not fit. Extracting derivatives from a finite truncated expansion has intrinsic limits: the spectral operators require a smooth density (a bare nuclear cusp breaks them, but pseudopotential densities are smooth) and the full window with ~ 40 channels; the far tail degrades where the finite-difference reference is itself noisy.

Appendix F: The exchange-hole functional: full derivation

We reconstruct the functional from the exact hole, flagging each degree of freedom (DOF) that is dropped or constrained, so the model’s location relative to the exact functional is explicit.

Exact starting point. The exchange energy is exactly Eq. (10) with $n_x(r_0, \mathbf{u}) = -|\rho_1(r_0, r_0 + \mathbf{u})|^2 / \rho(r_0)$, ρ_1 the one-particle density matrix. The hole obeys the sum rule $\int d^3u n_x = -1$, the on-top value $n_x(r_0, \mathbf{0}) = -\frac{1}{2} \rho(r_0)$, and $n_x \leq 0$. The full object is a two-point function with arbitrary angular dependence in $\mathbf{u}$.

DOF #1 — convolutional ansatz. We factor the hole into the actual neighboring density, a single universal shape S , and two local scalars: $n_x(r_0, \mathbf{u}) = -\rho(r_0 + \mathbf{u}) S(\zeta(r_0)u) / \mathcal{N}(r_0)$. The neighboring-density factor retains genuine non-local information; the *shape* is collapsed to one function plus a local inverse-length scale

ζ and a normalization $\mathcal{N}$.

DOF #2 — monopole projection (decisive). Inserting the ansatz into Eq. (10), the $1/u$ weight selects the spherical average of $\rho(r_0 + \mathbf{u})$, so only the $\ell = 0$ multipole survives (App. D): $\bar{\rho}(r_0, u) = A_0 \bar{\rho}_{\sigma, \text{safe}} \sum_n \varrho_{n00}(r_0) R_n(u)$. The model is intrinsically a spherically symmetric hole; every $\ell \geq 1$ hole multipole is dropped here. This is the largest DOF loss and is the source of the residual open-shell error.

Closed kernel (no DOF lost). With S fixed, the u -integrals precompute into the envelope tables $B_n(\zeta) = \int_0^\infty S(\zeta u) R_n(u) u^2 du$ and $C_n(\zeta) = \int_0^\infty S(\zeta u) R_n(u) u du$, the envelope-weighted charge and Coulomb moments. The number enclosed and the self-Coulomb of the enclosed density are linear contractions $Q_S(\zeta) = \mathbf{B}(\zeta) \cdot \boldsymbol{\varrho}_0$ and $\Phi_S(\zeta) = \mathbf{C}(\zeta) \cdot \boldsymbol{\varrho}_0$, giving the boxed self-energy $\varepsilon_x = -\frac{1}{2} \Phi_S / Q_S$: exchange per electron is $-\frac{1}{2}$ the self-Coulomb of the enclosed density divided by the number enclosed.

DOF #3 — closing the scale, and the exact limits. One scalar per point remains, fixed by the on-top sum rule $Q_S(\zeta) = 2$ (a spin-paired pair). This is where the limits enter:

- *HEG/LDA.* Uniform $\rho \Rightarrow \bar{\rho}$ constant; with the HEG hole $S(x) = [3j_1(x)/x]^2 = [j_0(x) + j_2(x)]^2$, $Q_S = 2 \Rightarrow \zeta = k_F$ and $\varepsilon_x \rightarrow \varepsilon_x^{\text{LDA}}$. Exact by construction; here S is frozen to the HEG shape.
- *Fermi-Amaldi.* Low density / one electron $\Rightarrow Q_S < 2$, giving the self-interaction-corrected form with the full enclosed density — exact for H/He.

The “2” is a spin assumption (a paired pair); the open-shell case is handled by the spin decomposition of App. H.

Recovering the gradient expansion. Inhomogeneity is returned to the shape through the $\ell = 1$ SIMPLE channel, whose squared distance from the uniform gas is the reduced gradient s^2 [Eq. (8)]. In the production functional this enters through the kernel fixed-point map (App. G): a single $\ell = 1$ kernel node has its amplitude fixed so the small- s slope reproduces the exact second-order coefficient $\mu = 10/81$ — with no explicit gradient term and no fitted parameter, and parity automatic since the coordinate is s^2 . A plane-wave check confirms the mechanism: perturbing the uniform gas with a density wave $\rho_0(1 + \alpha \cos kz)$ (analytic monopole, operator-free) shows the bare monopole hole is *blind* to the gradient (its s^2 response vanishes at the pure-gradient phase), so the $\ell = 1$ channel supplies the *entire* second-order term, recovering $\mu = 10/81$ as $\rho \rightarrow \text{HEG}$. The expansion is stopped at second order: only the $\ell = 1$ gradient is carried, whose adjoint grows as k_n^1 and is well-conditioned, whereas a fourth-order term would need the stiff $\ell = 0$ Laplacian channel (App. E).

The DOF ledger and the frontier.

DOF	where	restored?
dropped/constrained		
arbitrary radial profile	convolution	partly
$\rightarrow$ one $S(\zeta u)$	ansatz	
all $\ell \geq 1$ hole multipoles	monopole projection	pro- no
envelope shape to HEG	frozen $Q_S = 2$ closure	by kernel map
local scale ζ (paired)	+ spin on-top rule	no

The unremoved residual — a $\sim 10\%$ over-binding on strongly open, partially-filled subshells — is DOF #2: that exchange is carried by the $\ell \geq 1$ hole multipoles, and no deformation of a spherically symmetric envelope can supply them. The natural extension is an explicit multipole hole $n_x(r_0, \mathbf{u}) = \sum_\ell n_x^{(\ell)}(r_0, u) P_\ell(\hat{\mathbf{u}} \cdot \hat{\mathbf{r}}_0)$ using the $\ell \geq 1$ SIMPLE projections as genuine hole channels.

Appendix G: Kernel-mapped fixed-point hole: the GEA limit from the kernel

The production functional keeps the *direct* hole expansion $\tilde{n}(u) = -\frac{1}{2} \rho_0 \sum_n \tilde{\varrho}_n R_n^{(1)}(u/R_{\text{ad}})$ (with the on-top amplitude $-\frac{1}{2} \rho_0$ factored out so $\tilde{\varrho}$ is dimensionless) and produces its coefficients by a kernel interpolation over reference holes, with the gradient limit emerging from the kernel itself rather than an added term.

The energy is a single moment. The $1/u$ weight collapses the energy *exactly* onto the Coulomb moments $C_{1,n} = \int_0^1 R_n^{(1)}(t) t dt$: $\varepsilon_x = -\pi \rho_0 R_{\text{ad}}^2 (\tilde{\varrho} \cdot \mathbf{C}_1)$. Only the $\mathbf{C}_1$ -projection of $\tilde{\varrho}$ reaches the energy; orthogonal components carry hole shape only. Using the scale-free identity $R_{\text{ad}}^2 \rho_0^{2/3} = \xi^{*2} / (3\pi^2)^{2/3}$, the enhancement is a *linear functional of one moment* of the shape deviation $\delta \tilde{\varrho} = \tilde{\varrho} - \tilde{\varrho}_{\text{LDA}}$,

$$F_x - 1 = \kappa (\delta \tilde{\varrho} \cdot \mathbf{C}_1), \quad \kappa = \frac{\pi \xi^{*2}}{(3\pi^2)^{2/3} |C_{\text{LDA}}|}. \quad (\text{G1})$$

Kernel $\rightarrow$ moment. Let $\delta \tilde{\varrho}(x) = \sum_k \mathcal{K}(x, x_k) c_k$, $c = \mathbf{K}^{-1} \Delta$, be a general kernel interpolant over fixed points $\{x_k\}$ at per- ℓ squared distances $x = (D_0, D_1, D_2)$, $D_1 = s^2$, with node deviations Δ_k . Projecting on $\mathbf{C}_1$ collapses the vector interpolation to a *scalar* interpolation of the node moments $\mu_k = \Delta_k \cdot \mathbf{C}_1$: $\delta \tilde{\varrho}(x) \cdot \mathbf{C}_1 = \sum_k \mathcal{K}(x, x_k) m_k$ with $m = \mathbf{K}^{-1} \mu$. A HEG node at $x = 0$, $\mu = 0$, pins $F_x(0) = 1$ (LDA exact); because $\mathcal{K}$ is smooth in $D_1 = s^2$, $F_x - 1$ is analytic in s^2 , hence even in $\nabla \rho$ — parity is automatic, with no spurious linear-in-gradient term.

GEA2 from the kernel slope. The second-order limit is the $\ell = 1$ slope at the origin, a single linear condition on the node moments,

$$\left. \frac{\partial(F_x - 1)}{\partial D_1} \right|_{x=0} = \kappa \sum_k \partial_{D_1} \mathcal{K}(0, x_k) m_k \stackrel{!}{=} \frac{10}{81}. \quad (\text{G2})$$

For a Gaussian RBF $\mathcal{K} = e^{-\|x-x_k\|^2/2\ell^2}$ one has $\partial_{D_1}\mathcal{K}(0, x_k) = e^{-\|x_k\|^2/2\ell^2} D_1^k/\ell^2$; with the minimal HEG+GEA node pair $(x_G = (0, D_G, 0), \Delta_G = c_G \delta_{\text{GEA}})$ and the one-time response identity $\kappa(\delta_{\text{GEA}} \cdot \mathbf{C}_1) = R/2$, Eq. (G2) reduces to

$$\frac{c_G D_G \phi_G}{\ell^2(1 - \phi_G^2)} = \frac{20}{81R}, \quad \phi_G = e^{-D_G^2/2\ell^2}, \quad (\text{G3})$$

fixing the GEA-node amplitude c_G . With bulk reference nodes present, the same scalar condition (G2) — linear in c_G — is solved over the full node set, so the slope is hit exactly regardless of the references.

Why bulk references do not spoil the limit. GEA2 is a statement only of the slowly-varying limit, where every SIMPLE distance is small. A reference atom is *strongly* inhomogeneous, at finite $\|x_k\|$: it shifts the leading coefficient in (G2) by a constant that is absorbed in the calibration of c_G , while its uncontrolled effect is pushed to $O(s^4)$ — exactly the fourth-order GEA that is dropped. Along a slowly-varying ramp the monopole distance is higher order, $D_0 \ll D_1 = s^2$, so $(F_x - 1)/s^2 \rightarrow 10/81$. The HEG node keeps LDA exact; references far in feature space leave both limits untouched ($\delta\tilde{\rho} \rightarrow 0$) and shape only the strongly inhomogeneous interior.

Appendix H: Spin conventions

Exchange acts within each spin channel, so the spin-resolved energy follows from the spin-scaling relation $E_x[\rho_\uparrow, \rho_\downarrow] = \frac{1}{2}(E_x[2\rho_\uparrow] + E_x[2\rho_\downarrow])$. For a spin-paired system $\rho_\uparrow = \rho_\downarrow = \rho/2$, this reduces to the total-density expressions used in the main text, with the on-top count of 2. For open shells we evaluate the spin-restricted functional on $2\rho_\sigma$ in each channel and combine; the Hartree–Fock/OEP reference used for comparison is itself spin-restricted, which under-counts open shells, so the unrestricted-exact correction is applied when comparing.

Appendix I: Self-consistent potential and numerical details

Convolutional adjoint potential. Each monopole feature $q_{n00} = c'_{n00}/(A_0\bar{\rho}_{\sigma,\text{safe}})$ [Eq. (7)] is fixed by a *raw* coefficient c'_{n00} — the convolution of the density with a fixed kernel,

$$c'_{n00}(\mathbf{r}) = \int K_n(\mathbf{r}' - \mathbf{r}) \rho(\mathbf{r}') d^3r' = (K_n * \rho)(\mathbf{r}), \quad (\text{I1})$$

so that $\delta c'_{n00}(\mathbf{r}')/\delta\rho(\mathbf{r}) = K_n(\mathbf{r} - \mathbf{r}')$ exactly (this linear convolution is the coefficient the implementation forms as $\text{op}\otimes\rho$; the dimensionless feature follows by the fixed local rescaling). With the exchange energy $E_x = \int \rho \varepsilon_x[\{c'_{n00}\}] d^3r$, the potential is the explicit variational

derivative, following Ref. [6],

$$\begin{aligned} v_x(\mathbf{r}) &= \frac{\delta E_x}{\delta\rho(\mathbf{r})} \\ &= \varepsilon_x(\mathbf{r}) + \sum_n \int d^3r' \rho(\mathbf{r}') \frac{\partial \varepsilon_x(\mathbf{r}')}{\partial c'_{n00}} K_n(\mathbf{r} - \mathbf{r}'). \end{aligned} \quad (\text{I2})$$

The second term is itself a convolution over the kernel: each coefficient's contribution is K_n applied to $\rho \partial \varepsilon_x / \partial c'_{n00}$. In the radial finite-element implementation this convolution is the transpose P_n^T of the forward stencil P_n , weighted by the quadrature measure $w = 4\pi r^2 w_{\text{quad}}$ (w_{quad} the radial quadrature weights),

$$v_x = \varepsilon_x + \sum_n \frac{P_n^T[w \rho \partial \varepsilon_x / \partial c'_{n00}]}{w}, \quad (\text{I3})$$

exact against finite differences of the energy.

On-top scale by enclosed-charge inversion. The on-top scale is *not* found by a nested self-consistent iteration. The S -weighted enclosed charge $Q_S(\zeta) = \mathbf{B}(\zeta) \cdot \mathbf{q}_0 = \int_0^\infty S(\zeta u) \bar{\rho}(r_0, u) 4\pi u^2 du$ uses the precomputed envelope table $B_n(\zeta)$, so for the fixed monopole coefficients $\mathbf{q}_0$ it is a single matrix–vector product on a ζ -grid. As $\zeta \rightarrow 0$, $S \rightarrow 1$ and Q_S approaches the full window charge; as $\zeta \rightarrow \infty$, $Q_S \rightarrow 0$. Because $Q_S(\zeta)$ is monotonic, $Q_S(\zeta) = 2$ has a unique root recovered by one-dimensional inversion of the precomputed curve — the identical fixed-stencil enclosed-charge machinery that solves $g(R_{\text{ad}}) = \xi^*$ for the adaptive radius (App. B). The solve is therefore pointwise, non-iterative, and vectorized over all grid points, adding no loop to the Kohn–Sham cycle. (Confirmed in the implementation: the self-consistent functional reproduces the direct-integral hole on the converged density.)

Adjoint conditioning and the second-order cutoff. The exchange-kernel response $\delta v_x/\delta\rho$ is dominated by the $\ell = 1$ gradient channel. The $\ell = 1$ adjoint grows only as k_n^1 and is well-conditioned, so the second-order functional converges self-consistently across all atoms tested with no auxiliary loop. A fourth-order Laplacian term would instead enter through the $\ell = 0$ spectral Laplacian, whose adjoint re-forms the singular k_n^2 kernel and is catastrophically ill-conditioned (App. E); attempts to stabilize it (per-channel forward, frozen-correction outer loops) did not hold across cutoffs. This is why the expansion is stopped at second order.

The $-1/r$ tail. The exact exchange potential decays as $-1/r$. The model potential is made to recover this by an energy-neutral gauge fix (subtracting the asymptotic constant) together with a low-density damping $f = \rho^2/(\rho^2 + \rho_c^2)$ (ρ_c a small damping density), which restores the $-1/r$ tail and the correct Koopmans ionization potential without changing the energy.

Appendix J: Three-dimensional Cartesian-grid validation

The entire pipeline is run on genuine 3D Cartesian lattices, with window coefficients by direct kernel sums, ball averages by partial-cell shell sums (linear-ramp weights normalized by their own sum, keeping the HEG limit exact), and the adaptive radius, transfer, and non-dimensionalization as in 1D. Test densities are pseudo-diatomics: pseudo-H₂ (two analytic 1s atoms, $\rho = e^{-2r}/\pi$, $b = 1.40$ bohr; nuclear cusps) and pseudo-N₂ (two nitrogen pseudopotential valence densities from the radial DFT solver, GGA-PBE, $b = 2.074$ bohr; smooth). A quasi-exact continuum reference is built per-atom in an aligned frame and rotated with real Wigner- D matrices.

The descriptors converge as $\mathcal{O}(h^2)$ (0.1–1.2% at $h = 0.2$ bohr, 0.03–0.3% at $h = 0.1$); under 24 random rotations the per-channel deviation from the covariant prediction stays at the static-quadrature level ($\lesssim 8 \times 10^{-3}$) and the rotation-invariant power-spectrum spread is only $2\text{--}8 \times 10^{-4}$; symmetry-forbidden odd- ℓ channels at the bond midpoint vanish to machine precision ($\lesssim 2 \times 10^{-16}$); and a sub-grid registration sweep stays at $\sim 1.0 \times 10^{-3}$. The lattice scale-invariance test tracks the 1D radial behavior with the same two breakdown mechanisms. Thus ~ 0.2 bohr grid spacings, typical of real-space pseudopotential codes, are sufficient.

Appendix K: Practical limits of the SIMPLE features on real-space grids

The validation above establishes that the descriptors are convergent and covariant; this appendix maps the *practical* limits — where each feature family, used as a functional ingredient, stays invariant on a realistic grid and where it fails. We stress four ingredients through $\ell = 3$ — the reduced gradient s , the reduced Laplacian q , the power spectrum $P_{n\ell}$ (40 entries), and the bispectrum (all triangle triples, 1.3×10^4 entries) — on five superposition densities built from single-atom radial densities: pseudo-H₂, pseudo-N₂, pseudo-H₂O (O+2H at 0.96 Å, 104.5°), and FCC bulk Al and Pt (clusters filling the window). Everything is run at the fixed production setting ($R_c = 6$ bohr, $n_{\text{out}} = 10$, $n_{\text{in}} = 20$, so $\Lambda_{\text{max}} = n_{\text{in}}/n_{\text{out}} = 2$); the realistic spacings are $h = 0.1, 0.2, 0.3$ bohr; the ground truth is the analytic continuum-quadrature reference (for $\tilde{P}$ and the bispectrum) and a finite-difference gradient/Laplacian of the analytic density (for s, q). We flag any relative error $> 10\%$ whose feature magnitude is above a per-group floor (a large relative error on a near-zero feature carries no information). The four tests are translation (sub-grid registration), rotation (random rigid rotations), uniform scaling, and grid-spacing sensitivity; the script `invariant_stress_test.py` reproduces every number and figure.

Rotation and translation are robust (except q). Across 8 random rotations the power spectrum, bispec-

trum, and reduced gradient s hold to $\lesssim 1\%$ at $h = 0.2$ for every system and center (median $\ll 1\%$) — the grid breaks rotational symmetry only at the static-quadrature level. Under sub-grid registration the deviation stays $\lesssim 1\%$ except at the bare pseudo-H₂ nuclear cusp, where it reaches $\sim 10\text{--}12\%$ (the cusp is the hardest case for any uniform grid; pseudopotential and bulk densities are smooth and stay $\lesssim 1\%$). The reduced Laplacian q is the exception: it is invariant in the mean, but where its magnitude is small (e.g. the off-axis pseudo-H₂ center) the noise in its reconstruction produces large fractional excursions under both rotation and translation ($\sim 70\text{--}90\%$), already a warning that q is not a usable ingredient.

Grid-spacing: $\tilde{P}$ and the bispectrum converge; q is a reconstruction problem, not a grid problem. The power spectrum and bispectrum converge as $\mathcal{O}(h^2)$ toward the continuum (Fig. 8), reaching the 10% flag only for the cusped pseudo-H₂ at the coarsest $h = 0.3$; at $h = 0.2$ all systems are within $\sim 3\%$ and the smooth ones within $\sim 1\%$. The *grid-discretization* error of s and q is likewise small. Their limitation is instead the *reconstruction* fidelity (reconstructed feature versus the exact derivative): s is faithful in molecules ($\leq$ few percent for N₂, H₂O), but q — the $\ell = 0$ Bessel-eigenvalue contraction — is unreliable in dense environments, off by tens to hundreds of times at the bulk metal sites ($\sim 30\times$ and $\sim 160\times$ for the Al site and octahedral interstitial, $\sim 400\times$ for the Pt octahedral interstitial). This is the practical confirmation that the dimensionless Laplacian cannot be carried as a functional ingredient (Sec. IB 1, App. E); the high-symmetry bulk sites are inversion centers where $\nabla\rho \equiv 0$, so s itself vanishes there and its relative error is not informative, whereas q is genuinely large and genuinely wrong.

Uniform scaling: usable within Λ_{max} , resolution-gated, not a clamp. Restricting the scaling to within the theoretical range $\lambda \leq \Lambda_{\text{max}} = 2$ (set by the production $n_{\text{in}} = 20$, $n_{\text{out}} = 10$), the power spectrum and bispectrum stay near 1% and the reduced gradient near 10% (median $\sim 7\%$; Fig. 9); q is again the worst. The residual scale deviation is *not* the adaptive-radius clamp — across every system and scaling here R_{ad} stays well below R_c (0.3–3 bohr; never clamped) — nor a grid artifact (the continuum-quadrature features show the same deviation). It is the *inner-channel resolution* bound of the scale transform, $k^*\lambda \lesssim n_{\text{in}}\pi/R_c$ (Sec. IA 2): the transfer that reconstructs the dilated-window features needs enough inner channels to represent the dilated sampling. The reduced gradient is the most demanding because its slope weights $g_n^{(1)} \propto k_n$ emphasize the highest channels, so it converges last; raising n_{in} tightens the whole set systematically (Fig. 10: the descriptor-vector deviation at $\lambda = 1.67$ falls from $\sim 19\%$ at $n_{\text{in}} = 16$ through $\sim 13\%$ at the production $n_{\text{in}} = 20$ to $\sim 1\%$ at $n_{\text{in}} = 32$ and $\sim 0.2\%$ at $n_{\text{in}} = 48$). Scale invariance is thus exact in principle and a controllable resolution/cost trade-off in practice: a target dilation range Λ is bought by fixing $n_{\text{in}} \gtrsim n_{\text{out}}\Lambda$ (App. L).

Summary. At the production setting the power spectrum, the bispectrum, and the reduced gradient s are practically usable under all four operations — rotation and translation ($\lesssim 1\%$), grid refinement ($\mathcal{O}(h^2)$), and uniform scaling within $\Lambda_{\max}$ ($\tilde{P}$ /bispectrum $\sim 1\%$, $s \lesssim 10\%$, tightening with n_{in}). The reduced Laplacian q fails on multiple counts — rotation/translation noise where it is small, and a reconstruction against the exact $\nabla^2 \rho$ that is off by tens to hundreds of times in bulk metals (its $\ell = 0$ eigenvalue sum is ill-conditioned) — so it is excluded, consistent with the q -free, second-order functional of the main text.

Appendix L: Parameter selection: window, resolution, scale range, and cost

The limits of App. K fix the few free parameters of the descriptor — the window R_c , the grid spacing h , the inner- and exposed-channel counts $n_{\text{in}}, n_{\text{out}}$, and the angular cutoff $\ell_{\max}$ — through two requirements: a target energy accuracy (which sets R_c and n_{out}) and a target scale-invariance range (which couples h , n_{in} , n_{out} , and the cost). We make both quantitative and adopt a single production setting, $R_c = 6$ bohr and $n_{\text{out}} = 10$, used in the rest of the paper.

Scale range, resolution, and cost are one relation. On a grid of spacing h the inner channels resolvable below the Nyquist wavevector π/h number $n_{\text{in}} = R_c/h$ (the highest channel $k_{n_{\text{in}}} = n_{\text{in}}\pi/R_c = \pi/h$ sits exactly at Nyquist). The scale transform reconstructs the dilated-window descriptors only while the dilated sampling stays within that resolution, $k^*\lambda \lesssim n_{\text{in}}\pi/R_c$ with $k^* \simeq n_{\text{out}}\pi/R_c$ the highest *exposed* wavevector, i.e. the features stay scale-invariant up to a dilation

$$\Lambda_{\max} \simeq \frac{n_{\text{in}}}{n_{\text{out}}} = \frac{R_c}{h n_{\text{out}}}. \quad (\text{L1})$$

This is confirmed directly (Fig. 13a): at fixed $n_{\text{out}} = 10$, the descriptor-vector deviation $D(\lambda)$ stays $\lesssim 1\%$ until λ approaches $\Lambda_{\max} = n_{\text{in}}/n_{\text{out}}$ and rises beyond it, with the usable range growing linearly in n_{in} ($\Lambda_{\max} \approx 2.0, 3.0, 6.0$ at $h = 0.3, 0.2, 0.1$ bohr). The cost is set by the same n_{in} : each descriptor is one fixed-stencil convolution per radial channel and angular index, so

$$N_{\text{conv}} = n_{\text{in}}(\ell_{\max} + 1)^2 = \frac{R_c}{h}(\ell_{\max} + 1)^2, \quad (\text{L2})$$

the transfer to n_{out} being a cheap per-point matrix product. Eliminating h between Eqs. (L1)–(L2) gives the design rule directly: to be scale-invariant over a dilation Λ with n_{out} exposed channels through $\ell_{\max}$ requires

$$h \leq \frac{R_c}{n_{\text{out}} \Lambda}, \quad N_{\text{conv}} \geq n_{\text{out}} \Lambda (\ell_{\max} + 1)^2. \quad (\text{L3})$$

For $R_c = 6$, $n_{\text{out}} = 10$, $\ell_{\max} = 3$ (so $(\ell_{\max} + 1)^2 = 16$) this is $N_{\text{conv}} = 320, 480, 960$ convolutions at $h =$

$0.3, 0.2, 0.1$ bohr (set by n_{in} , not n_{out}), buying $\Lambda_{\max} \approx 2.0, 3.0, 6.0$ (Fig. 13b) — the coarsest practical grid ($h = 0.3$) already covers a $2.0\times$ linear ($\sim 8\times$ density) dilation at 320 convolutions, and halving h buys proportionally more range at proportionally more cost.

The cost is set by the scale range, not the grid. A subtlety worth making explicit: n_{in} is a *design choice* — the number of inner channels one elects to compute — bounded above only by the grid through $n_{\text{in}} \leq R_c/h$. The sequence $N_{\text{conv}} = 320, 480, 960$ above corresponds to spending the *full* grid resolution $n_{\text{in}} = R_c/h$ at each h , which is why it grows as the grid refines (and buys the larger $\Lambda_{\max} = 2, 3, 6$). But for a *fixed* target scale range one simply fixes $n_{\text{in}} = n_{\text{out}}\Lambda$; the convolution count $N_{\text{conv}} = n_{\text{out}}\Lambda(\ell_{\max} + 1)^2$ is then independent of h , and the grid need only be fine enough to resolve it, $h \leq R_c/n_{\text{in}}$. Numerically, holding $n_{\text{in}} = 2n_{\text{out}} = 20$ fixed (so $N_{\text{conv}} = 320$ at $\ell_{\max} = 3$, for the full $\Lambda_{\max} = 2$): the scale deviation $D(\lambda)$ is essentially *h -independent* ($D(\lambda=2) \approx 0.035$ at $h = 0.3, 0.2, 0.1, 0.05$ bohr), while refining h only sharpens those same channels (their deviation from the continuum falls $0.5\% \rightarrow 0.01\%$ from $h = 0.3$ to 0.05). So a finer grid does *not* require more convolutions for a given scale range — it improves accuracy at fixed cost. Conversely, extending Λ means raising n_{in} (more convolutions) and refining h to resolve it; reducing the angular cutoff is the other lever, $N_{\text{conv}} = n_{\text{in}}(\ell_{\max} + 1)^2 = 320, 180, 80$ at $n_{\text{in}} = 20$ for $\ell_{\max} = 3, 2, 1$. The full $\ell \leq 3$ feature set at $\Lambda = 2$ is thus a fixed 320 convolutions on any grid $h \leq 0.3$ bohr.

An energy scale fixes R_c and n_{out} . To anchor the two remaining counts to a physical tolerance we evaluate the Hartree self-energy of the hydrogen $1s$ density ($\rho = e^{-2r}/\pi$, exact $J = 5/16$ Ha) through the SIMPLE windowed Coulomb projection [Eq. (9)], $V_H(r_0) = A_0 \bar{\rho}_{\sigma, \text{safe}} \sum_n C_n \varrho_{n00}$ with the monopole coefficients ϱ_{n00} in an n_{out} -channel basis and Coulomb moments $C_n = 4\pi \int_0^{R_c} R_n(u) u du$. Two truncations control the error (Fig. 11): the window R_c (the Coulomb tail beyond it is dropped) and n_{out} (the radial-basis resolution of the enclosed density). The window floor reaches the milli-Hartree scale only at $R_c \gtrsim 6$ bohr — -1.7 mHa at $R_c = 5$, **-0.37 mHa at $R_c = 6$** , -0.01 mHa at $R_c = 8$ — so $R_c = 6$ is the smallest window consistent with sub-milli-Hartree Coulomb energetics, and no n_{out} can rescue a shorter window ($R_c = 4$ plateaus at -7.9 mHa). At $R_c = 6$ the radial-basis error falls below 1 mHa already by $n_{\text{out}} \approx 8$ and effectively reaches the window floor by $n_{\text{out}} \approx 10$ (-0.49 mHa at $n_{\text{out}} = 10$; further channels gain little, the residual being the tail). Hence $R_c = 6$ bohr and $n_{\text{out}} = 10$ deliver the Coulomb energy to ~ 0.5 mHa with margin — the production setting.

The energy floor on a practical grid. The -0.5 mHa figure above is the continuum value of the projection; on a real Cartesian grid it acquires a discretization error that depends on the spacing h and on where the nucleus sits relative to the lattice (sub-grid registration). Evaluating the same H Hartree energy on a 3D grid (the windowed-

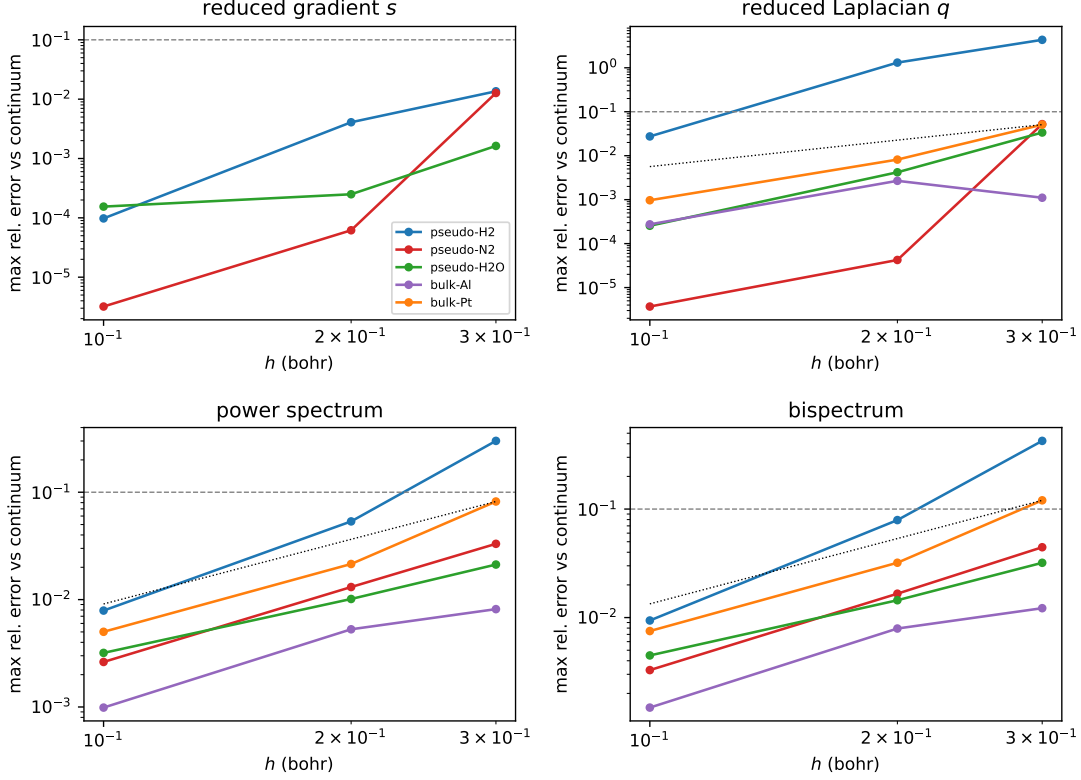
(4) Grid-spacing sensitivity: grid vs continuum ($R_c=6$ bohr); dashed = 10% flag, dotted = $\mathcal{O}(h^2)$ 

FIG. 8. Grid-spacing sensitivity of the four feature families: maximum relative error of the gridded features against the continuum reference versus spacing h , at $R_c = 6$ bohr, for the five systems (max over evaluation centers). Dashed line, the 10% flag; dotted, an $\mathcal{O}(h^2)$ guide. The power spectrum and bispectrum converge as $\mathcal{O}(h^2)$; only the cusped pseudo- H_2 approaches the flag at $h = 0.3$. (This is grid-discretization error; the reconstruction error of s and especially q in dense environments, discussed in the text, is a separate and larger effect.)

Coulomb kernel convolved with the gridded density) for four registrations — nucleus on a lattice point, on a face center, on the body center, and a generic offset — shows the error fanning out from the continuum floor as h coarsens (Fig. 12): all registrations agree to 0.02 mHa at $h = 0.1$, spread to 0.33 mHa at $h = 0.2$ (all within $[-0.55, -0.22]$ mHa) and 0.80 mHa at $h = 0.25$, and reach a 1.66 mHa spread at $h = 0.3$ where the best/worst registrations (+0.86/−0.80 mHa) just brush the ± 1 mHa band. The on-site placement overshoots and the body-centered undershoots — a grid-quadrature aliasing that averages out only as $h \rightarrow 0$. So the 1 mHa Coulomb tolerance holds robustly and registration-independently for $h \lesssim 0.2\text{--}0.25$ bohr; $h = 0.3$ bohr is the practical edge, where the translation-induced variation alone approaches

the tolerance.

Recommended setting. $R_c = 6$ bohr, $n_{\text{out}} = 10$, $\ell_{\text{max}} = 3$: Hartree energetics converged to ~ 0.5 mHa (continuum), holding to < 1 mHa on grids with $h \lesssim 0.2\text{--}0.25$ bohr for any registration (Fig. 12); $N_\ell = 10$ monopole channels for the spherically symmetric hole functional; and (Eq. (L3)) scale invariance over $\Lambda_{\text{max}} \approx 2.0$ at the coarsest $h = 0.3$ bohr, rising to ≈ 6 at $h = 0.1$ bohr, at 320–960 convolutions for the full $\ell \leq 3$ feature set. The reduced gradient s and the power spectrum/bispectrum are well within their flagged tolerances at these settings (App. K); the reduced Laplacian q is excluded on the stability grounds established there and in App. E.

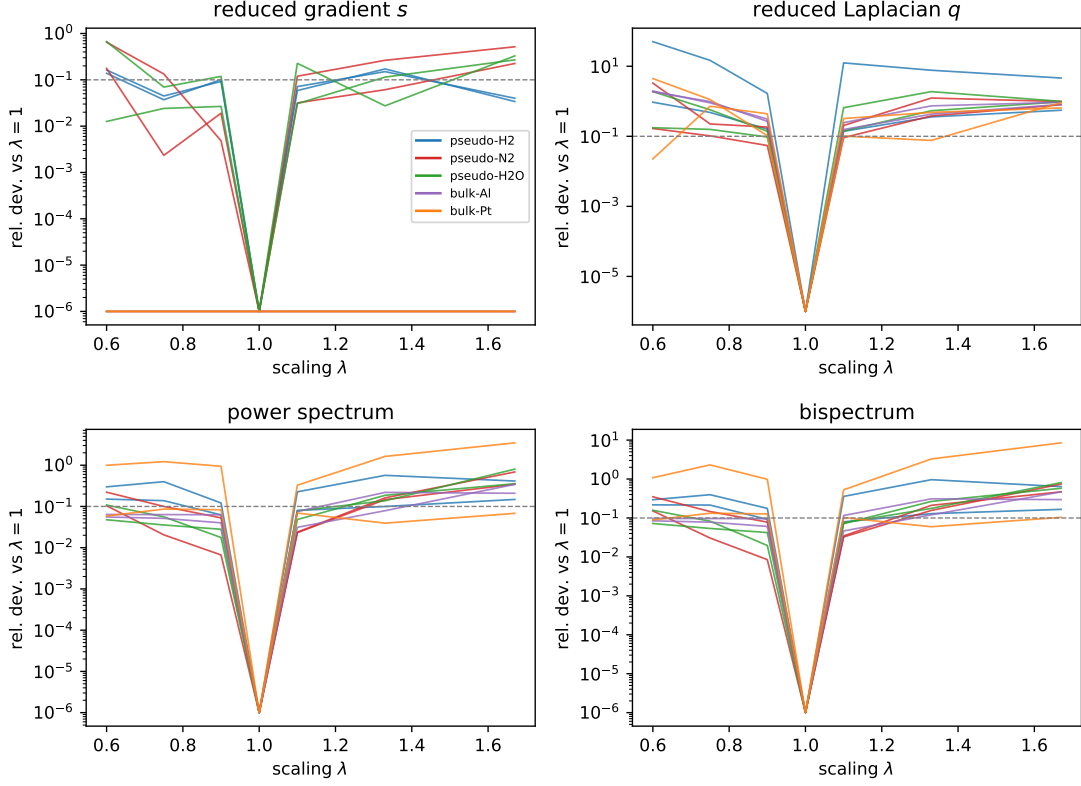
(3) Scale invariance, $h=0.2$ bohr; dashed = 10% flag

FIG. 9. Scale invariance at $h = 0.2$ bohr, production $n_{\text{in}} = 20$: relative deviation of each feature family from its $\lambda = 1$ value under uniform scaling $\rho \rightarrow \lambda^3 \rho(\lambda r)$, one curve per system and evaluation center (deviations floored at 10^{-6} for the log axis; dashed line, the 10% flag). The power spectrum, bispectrum, and reduced gradient grow gradually with $|\lambda - 1|$ within $\Lambda_{\text{max}} = 2$, while the reduced Laplacian is largest. The deviation is the inner-channel resolution of the scale transform (Fig. 10), not the adaptive-radius clamp (never reached here: $R_{\text{ad}} < R_c$ throughout), and the per-entry maximum shown amplifies the smallest near-floor entries.

Scale invariance is set by the inner-channel count n_{in} (continuum, $R_c = 6$ bohr; R_{ad} unclamped); dotted = Fig.~3 continuum level

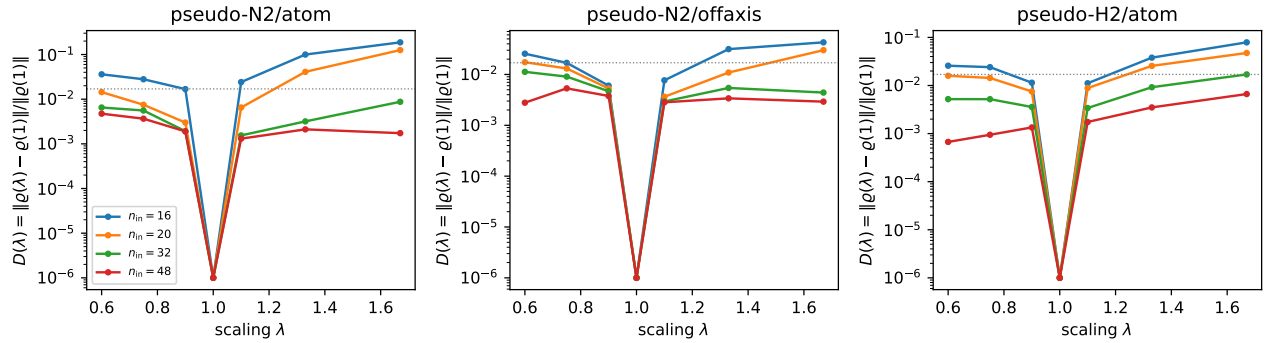


FIG. 10. Resolving the scale-invariance deviations of Fig. 9. Continuum-quadrature descriptor-vector deviation $D(\lambda) = \|\varrho(\lambda) - \varrho(1)\|/\|\varrho(1)\|$ under uniform scaling, for increasing inner-channel counts n_{in} , at $R_c = 6$ bohr (the adaptive radius is unclamped throughout). The deviation falls systematically with n_{in} — from $n_{\text{in}} = 16$ through the production $n_{\text{in}} = 20$ to $n_{\text{in}} = 48$ — collapsing below the Fig. 3 continuum level (dotted) once $n_{\text{in}} \gtrsim 32$. The scale non-invariance is therefore the documented inner-channel resolution bound $k^* \lambda \lesssim n_{\text{in}} \pi / R_c$, not a breakdown of the construction: with enough radial channels the transferred features are scale-invariant as the theory requires.

Hartree energy of H through the SIMPLE Coulomb projection: $R_c = 6$, $n_{\text{out}} = 12 \Rightarrow 0.4$ mHa

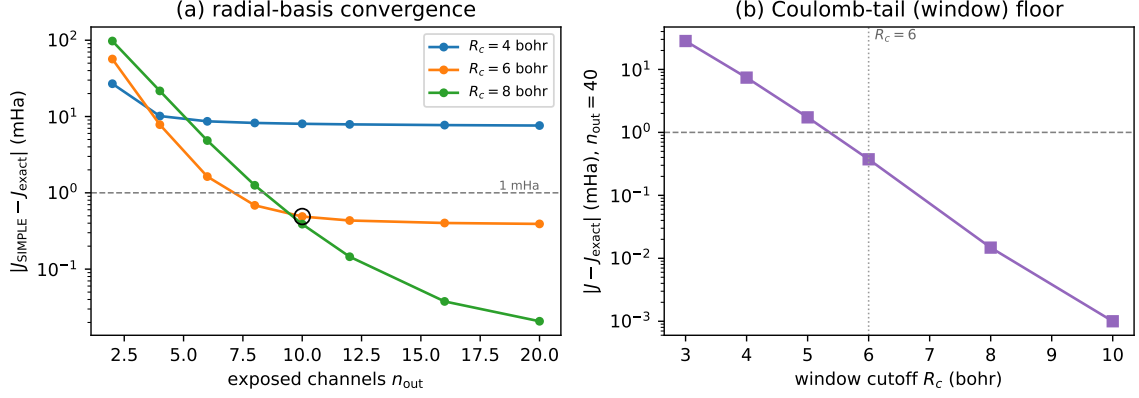


FIG. 11. Energy-scale calibration of R_c and n_{out} : the Hartree self-energy of the hydrogen $1s$ density through the SIMPLE windowed Coulomb projection [Eq. (9)], error versus the exact $5/16$ Ha. **(a)** radial-basis convergence versus n_{out} for several windows (circle: the adopted $R_c = 6, n_{\text{out}} = 10$ at -0.49 mHa); a short window ($R_c = 4$) is tail-limited and never reaches 1 mHa. **(b)** the n_{out} -converged window floor versus R_c , crossing 1 mHa at $R_c \approx 6$ bohr. Together they fix $R_c = 6, n_{\text{out}} = 10$.

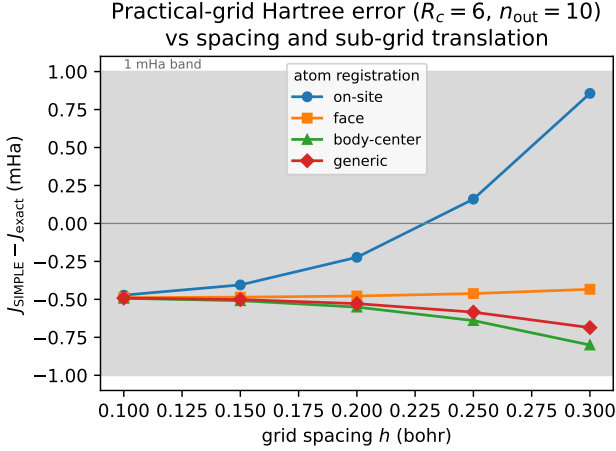


FIG. 12. The Hartree-energy tolerance on a practical grid: the H $1s$ Hartree energy through the SIMPLE windowed-Coulomb projection ($R_c = 6, n_{\text{out}} = 10$) evaluated on a 3D Cartesian grid, versus spacing h and the nucleus's sub-grid registration (on a lattice point, a face center, the body center, a generic offset). All registrations collapse to the -0.49 mHa continuum floor as $h \rightarrow 0$; the grid-discretization error fans out with h (on-site overshoots, body-centered undershoots), staying inside the ± 1 mHa band (shaded) for $h \lesssim 0.25$ bohr and reaching a 1.66 mHa registration spread at $h = 0.3$ bohr. The 1 mHa Coulomb tolerance is thus registration-robust for $h \lesssim 0.2$ – 0.25 bohr.

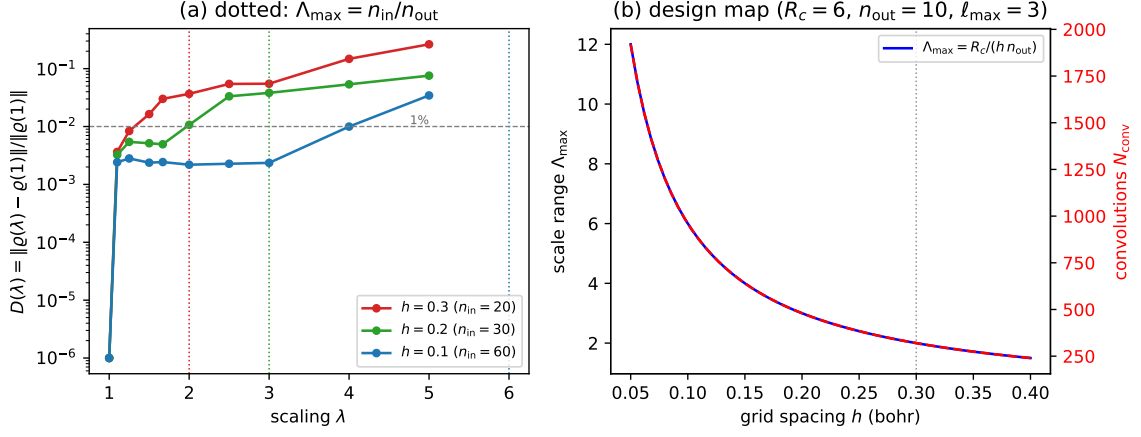


FIG. 13. Scale range versus resolution and cost at the production $n_{out} = 10$, $R_c = 6$ bohr. **(a)** descriptor-vector deviation $D(\lambda)$ under uniform scaling for $h = 0.3, 0.2, 0.1$ bohr ($n_{in} = R_c/h = 20, 30, 60$); dotted verticals mark the predicted $\Lambda_{max} = n_{in}/n_{out}$ [Eq. (L1)], which the 1% crossings track. **(b)** design map: the scale range $\Lambda_{max} = R_c/(h n_{out})$ and the convolution count $N_{conv} = (R_c/h)(\ell_{max} + 1)^2$ versus h (both $\propto 1/h$); the dotted line marks the coarsest practical $h = 0.3$ bohr.